



THE UNIVERSITY OF
SYDNEY

THE UNIVERSITY OF SYDNEY

SCHOOL OF AEROSPACE, MECHANICAL AND MECHATRONIC ENGINEERING

AERO3260: AERODYNAMICS 1

Flow Around a Circular Cylinder and a Two-Dimensional Aerofoil

LABORATORY REPORT 1

Lab Group: Wednesday 11:00 A

MEMBER	SID
Austin MARKWICK	520419773
Hamish IERINO	540750496
James EDWARDS	530474595
Unai ARCE ANDONEGUI	560779376

2 September 2026

Abstract

The experiments reported here were conducted at 11 am on 19 August 2026 in the Wind Tunnel Laboratory, Mechanical Engineering Building S173 Area 5, at the University of Sydney. Two low-speed wind tunnel experiments were performed to characterise the pressure distributions and aerodynamic forces on a circular cylinder and on a NACA 23012 aerofoil. Ambient conditions gave an air density of 1.214 kg/m^3 and a dynamic viscosity of $1.828 \times 10^{-5} \text{ kg/(m s)}$. Surface pressures on a 50 mm cylinder were recorded with a multi-tube manometer at Reynolds numbers of 2.0×10^4 and 3.0×10^4 , both in the sub-critical regime, and showed laminar separation near 80° with a flat base pressure in place of the recovery predicted by potential flow. Pressure integration gave a drag coefficient of 1.29 at the higher Reynolds number, within 8 % of published data, and the force transducer gave 1.06 once the support structure was subtracted. A NACA 23012 section of 175 mm chord was then tested at $\text{Re} = 1.4 \times 10^5$ over angles of attack from -10° to 20° using a Scanivalve pressure scanner and a wake rake. The measured lift curve has a slope of 6.0/rad and a zero-lift angle of -2.0° , the section stalls between 16° and 18° with a maximum lift coefficient of 1.51, and the wake momentum integral gives a profile drag coefficient of 0.021 near zero angle of attack. The two drag methods agree where form drag dominates and diverge where skin friction, tap coverage or rake width limit one of them, which the report quantifies and explains.

Contents

List of Figures	v
List of Tables	vi
1 Introduction	1
2 Methods	1
3 Results and Discussion	2
3.1 Ambient Conditions	2
3.2 Circular Cylinder	2
3.3 NACA 23012 Aerofoil	4
4 Conclusion	6
References	7
Appendices	8
A Raw Data	8
A.1 Test Section Photographs	8
A.2 Cylinder Pressure Tap Mapping	8
A.3 Cylinder Force Coefficients from the Tap Pressures	9
A.4 Numerical Evaluation of the Cylinder Integrals	10
A.5 Cylinder Drag Coefficient against Reynolds Number	11
A.6 Flow Regimes of a Smooth Circular Cylinder	12
A.7 Cylinder Manometer and Force Readings	12
A.8 Error Bars	13
A.9 Aerofoil Pressure Tap Locations	13
A.10 NACA 23012 Section Definition	13
A.11 Aerofoil Pressure Distributions by Angle of Attack	14
A.12 Aerofoil Surface and Wake Pressures	14
A.13 Manometer Datasheet	16
A.14 Pressure Scanner Datasheet	18

B	Theoretical Validation with XFOIL	20
B.1	Computational Setup	20
B.2	Theoretical and Overlay Polars	20
C	Analysis Code	22

List of Figures

1	Experimental arrangements, not to scale. (a) Cylinder rig in elevation, held between end plates on a strut carried by the force transducer. (b) The aerofoil from above, with the wake rake and the static taps 48 and 49. Photographs in Appendix A.1 show the apparatus used in the laboratory.	1
2	Measured surface pressure coefficient distribution at $Re = 2.0 \times 10^4$ and 3.0×10^4 against the potential-flow distribution. Error bars are the manometer reading resolution (Appendix A.8).	3
3	Cylinder wake from sub-critical to post-critical. Red marks the forward stagnation point, black is separation at the labelled θ , T the laminar-to-turbulent transition, and shading is the wake.	4
4	Surface pressure coefficient on the NACA 23012 at $Re = 1.4 \times 10^5$, suction plotting upward. Error bars are the r.m.s. fluctuation of C_p over the 2500 recorded frames (Appendix A.8).	5
5	Coefficients and drag polar at $Re = 1.4 \times 10^5$ from pressure integration, with the wake drag of Section 3.3.4 on the C_d panel.	5
6	Normalised wake profiles at each angle of attack, every trace offset and read vertically against its own gridline at $u/U_\infty = 1$. The shaded area is the velocity deficit, y is positive towards port 47, and the error bars are the r.m.s. fluctuation carried through Equation (3.12) (Appendix A.8).	6
7	The test section as set up, annotated. Both models span the tunnel vertically, the cylinder between end plates and the aerofoil between the tunnel walls, with the pressure tubing leaving through the floor.	8
8	Orientation of the pressure taps on the cylinder. Odd-numbered taps past port 1 sit below the horizontal axis and even-numbered taps above it, with θ measured from the front stagnation point.	9
9	Variation of the drag coefficient of a circular cylinder with Reynolds number, reproduced from figure 1 of the laboratory brief. Data compiled by Panton [6], as reproduced by Anderson [1].	11
10	The data of Figure 4 decomposed by angle of attack, with the same r.m.s. bars and the vertical axis inverted so suction plots upward.	14
11	Theoretical aerodynamic coefficients of the NACA 23012 aerofoil from XFOIL at $Re = 1.4 \times 10^5$.	20
12	Comparison of experimental pressure-integrated data (blue) and wake data (green) against theoretical XFOIL data (red).	21

List of Tables

1	Ambient conditions measured before testing (T , p_{atm}) and the air properties derived from them by Equations (3.1) and (3.2) (ρ , μ).	2
2	Cylinder force coefficients from pressure integration and from the corrected transducer. . . .	3
3	Manometer tube number against angular position on the cylinder.	8
4	Regimes of flow about a smooth circular cylinder in steady current, after Lienhard [14] as reproduced by Sumer and Fredsøe [15]. The boundaries are the ones usually quoted and vary between sources and with surface roughness. Figure 3 expands the last three rows, which span the sub-critical to post-critical range of Section 3.2.5.	12
5	Raw manometer column heights in mm of water. Tube 14 is connected to the tunnel static port.	12
6	Force transducer readings for the cylinder and support structure.	12
7	Chordwise positions of the NACA 23012 pressure taps, after tables 2 and 3 of the brief. Tubes 1 to 13 serve the lower surface and 14 to 26 the upper.	13
8	Mean surface tap pressures (Pa) at each angle of attack ($^\circ$). Ports 1 to 13 are the lower surface and 14 to 26 the upper surface.	15
9	Mean wake rake and tunnel static pressures (Pa) at each angle of attack ($^\circ$). Ports 33 to 47 are the rake totals, port 48 the tunnel-inlet static and port 49 the side-wall static.	16

1 Introduction

The aerodynamic character of a body is governed largely by whether its boundary layer stays attached. A circular cylinder is the canonical bluff body, its drag set almost entirely by the pressure imbalance across a separated wake and therefore strongly dependent on Reynolds number [1], while an aerofoil is streamlined to keep the flow attached until the stall. Three techniques are used: surface pressure integration, which sees form drag alone and only where the taps reach; a force balance, which senses the total load but also that on the support; and a wake rake, which returns profile drag from the momentum deficit [2]. The objectives were to determine the laboratory air properties, to measure the cylinder pressure distribution and compare it with the potential-flow result $C_p = 1 - 4 \sin^2 \theta$, to obtain the cylinder force coefficients independently by pressure integration and from the transducer, and to measure the pressure distributions, force and moment coefficients and wake profiles of a NACA 23012 aerofoil from -10° to 20° . The measurements placed the cylinder firmly in the sub-critical regime and carried the aerofoil's lift, drag and quarter-chord moment curves through the stall.

2 Methods

Both experiments were performed in the low-speed wind tunnel of the Wind Tunnel Laboratory, with the speed set from the tunnel controller and held until the reading was steady, in the arrangements sketched in Figure 1. The cylinder model has a diameter of 50 mm and an exposed length of 110 mm, carries end plates to keep the flow two-dimensional, and sits on a support strut of 25 mm diameter with 70 mm exposed. Thirteen surface taps spiral around the mid-span at 15° increments, alternating sides of the cylinder from the front stagnation line to the base (Appendix A.2). The aerofoil is a NACA 23012 section of 175 mm chord mounted vertically across nearly the full height of the test section, with thirteen taps on each surface at the chordwise locations of Appendix A.9 and short wool tufts taped on for flow visualisation.

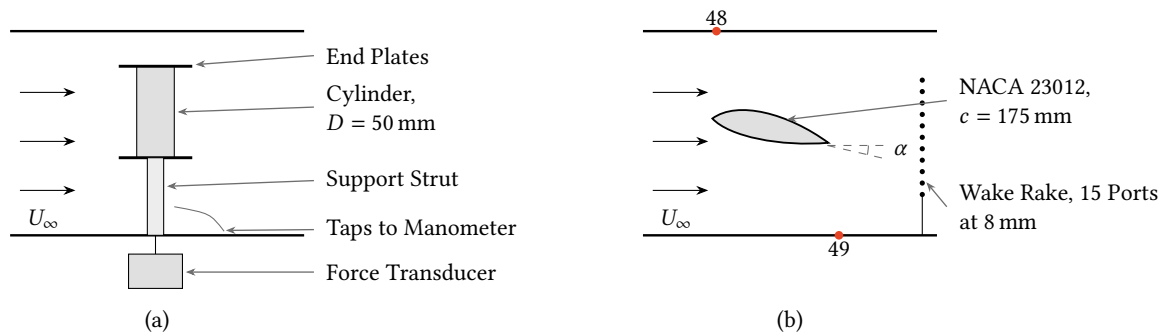


Figure 1: Experimental arrangements, not to scale. (a) Cylinder rig in elevation, held between end plates on a strut carried by the force transducer. (b) The aerofoil from above, with the wake rake and the static taps 48 and 49. Photographs in Appendix A.1 show the apparatus used in the laboratory.

Cylinder surface pressures were read from a sixteen-tube vertical water manometer graduated in millimetres and read to 0.5 mm, thirteen tubes serving the surface taps and one the tunnel static port, and the force on the cylinder and its support was measured by a three-component force transducer reporting lift, drag and pitching moment, zeroed after the model was installed so that only the aerodynamic load was recorded. Aerofoil surface pressures, the wake and the tunnel statics were recorded by a Scanivalve MPS 4264 miniature pressure scanner of four inch water range, which sampled every channel simultaneously for 2500 frames per run, with ports 1 to 26 on the surface taps, 33 to 47 on a wake rake of fifteen total-pressure tubes at 8 mm pitch behind the model, and 48 and 49 on the tunnel-inlet and side-wall statics. Ambient temperature and atmospheric pressure were read from the laboratory thermometer and barometer before testing began, giving the air properties used throughout Section 3.

With the ambient conditions and geometry recorded, the cylinder runs logged the manometer columns and zeroed-transducer forces at 6 and 9 m/s once the tunnel had stabilised. For the aerofoil the Scanivalve was zeroed with the tunnel off, which its two wind-off records confirm, and all channels were sampled at 12 m/s at each angle of attack from -10° to 20° in steps of 2° , with the tuft behaviour noted at each setting.

3 Results and Discussion

3.1 Ambient Conditions

The ambient temperature and atmospheric pressure were measured in the laboratory before the experiments began, giving $T = 23.0^\circ\text{C}$ and $p_{atm} = 1031.7\text{ hPa}$. Both were converted before use, to $T = 296.15\text{ K}$ and $p_{atm} = 103\,170\text{ Pa}$, since the ideal gas law requires absolute temperature. The density then follows from Equation (3.1) and the dynamic viscosity from Sutherland's law [3], Equation (3.2):

$$\rho = \frac{M_{air} p_{atm}}{R_u T} \quad (3.1)$$

$$\mu = \mu_0 \left(\frac{T}{T_0} \right)^{3/2} \frac{T_0 + S}{T + S} \quad (3.2)$$

The molar mass of dry air $M_{air} = 28.9645 \times 10^{-3}\text{ kg/mol}$ and the universal gas constant $R_u = 8.3145\text{ J/(mol K)}$ combine to the specific gas constant $R = R_u/M_{air} = 287.058\text{ J/(kg K)}$, so Equation (3.1) is equivalently $\rho = p_{atm}/(RT)$. In Equation (3.2) the reference dynamic viscosity is $\mu_0 = 1.716 \times 10^{-5}\text{ kg/(m s)}$ at $T_0 = 273.15\text{ K}$, and $S = 110.4\text{ K}$ is Sutherland's temperature. The resulting properties are given in Table 1 and are used in place of standard sea-level conditions throughout the report.

Table 1: Ambient conditions measured before testing (T , p_{atm}) and the air properties derived from them by Equations (3.1) and (3.2) (ρ , μ).

Temperature	Pressure	Density	Viscosity
T (K)	p_{atm} (Pa)	ρ (kg/m ³)	μ (kg/(m s))
296.15	103 170	1.214	1.828×10^{-5}

3.2 Circular Cylinder

3.2.1 Surface Pressure Distribution

Each tap reading (Appendix A.7) is referenced to the static port on tube 14. With the air density ρ of Section 3.1 and an assumed water density $\rho_w = 1\,000\text{ kg/m}^3$, the pressure coefficient is

$$C_p(\theta) = \frac{p(\theta) - p_\infty}{\frac{1}{2}\rho U_\infty^2} = \frac{\rho_w g [h_s - h(\theta)]}{\frac{1}{2}\rho U_\infty^2}, \quad (3.3)$$

where $h(\theta)$ and h_s are the tap and reference column heights and U_∞ the tunnel speed. The taps alternate sides around the cylinder (Appendix A.2), so readings are plotted against $|\theta|$ assuming a symmetric mean flow, and Figure 2 compares both runs with the potential-flow distribution $C_p = 1 - 4 \sin^2 \theta$.

Over the forward face, the 6 m/s run tracks the theory closely: every point out to 45° lies within 0.17 of it, inside the ± 0.22 uncertainty, with the zero crossing at 30° , as required. The falling pressure keeps the boundary layer thin and attached, so the inviscid solution holds, and the minimum of -3 corresponds to a surface speed of $2U_\infty$. The 9 m/s run has the same shape in this region, but is offset by about $+0.6$, with $C_p(0) = 1.60$ exceeding the expected stagnation $C_p = 1$, so the static reference was likely misread. Both level off between 60° and 75° and reach $C_p = -3.1$ and -0.8 at the base, because the adverse pressure gradient past the minimum stalls near-wall fluid already retarded by friction, separating the boundary layer at 82.4° and 81.7° , by the correlation of Jiang [4], $\theta_s = 78.8 + 505 \text{ Re}^{-1/2}$. The surface behind lies in a wake of nearly uniform pressure [5], whereas the theory recovers symmetrically to $+1$, leaving base gaps of 4.1 and 1.8 that are the form drag of Section 3.2.2, which the ideal distribution lacks by d'Alembert's paradox, the zero drag that potential flow predicts for any closed body [1].

3.2.2 Force Coefficients from Pressure Integration

For a cylinder, the lift and drag coefficients can be reduced to the pressure integrals $C_d = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta \, d\theta$ and $C_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta \, d\theta$, derived in Appendix A.3, with θ measured clockwise from the front stagnation

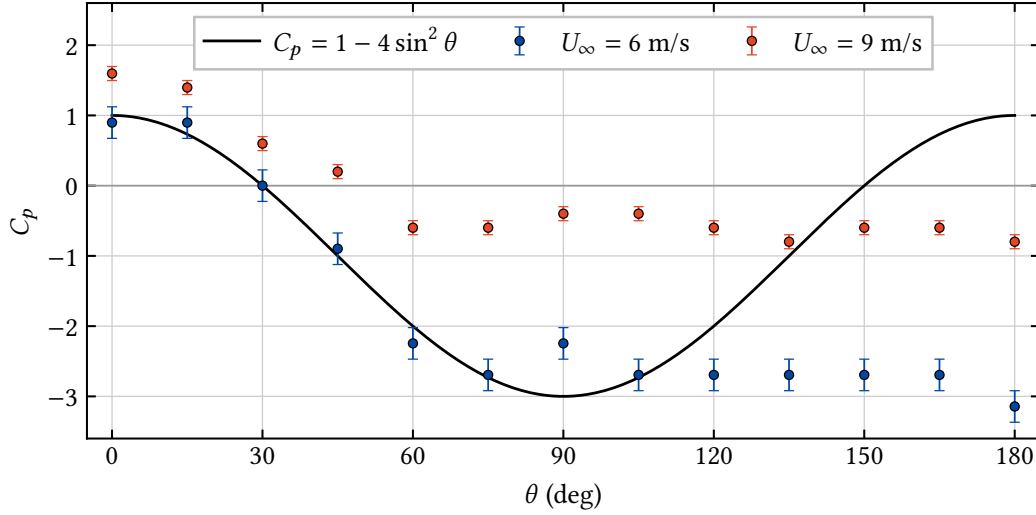


Figure 2: Measured surface pressure coefficient distribution at $Re = 2.0 \times 10^4$ and 3.0×10^4 against the potential-flow distribution. Error bars are the manometer reading resolution (Appendix A.8).

point. The thirteen pressure taps alternate sides (Appendix A.2) and close the contour, so both the lift and drag coefficients are evaluated over the cylinder surface via a composite trapezoidal rule (numerical integration) as,

$$C_d \approx \frac{1}{4} \sum_{i=1}^{13} (C_{p,i} \cos \theta_i + C_{p,i+1} \cos \theta_{i+1}) \Delta \theta_i, \quad (3.4)$$

$$C_l \approx -\frac{1}{4} \sum_{i=1}^{13} (C_{p,i} \sin \theta_i + C_{p,i+1} \sin \theta_{i+1}) \Delta \theta_i, \quad (3.5)$$

with $\Delta \theta_i = \theta_{i+1} - \theta_i$. At $Re = 2.0 \times 10^4$, this yields $C_d = 2.48$ and $C_l = +0.06$. At 3.0×10^4 , this yields $C_d = 1.29$ and $C_l = -0.05$, where drag is within 8 % of the sub-critical $C_d \approx 1.2$ of Panton [6].

3.2.3 Force Coefficients from the Force Transducer

The transducer senses the combined load on the cylinder and its support. The support is itself a circular cylinder at $Re_s = 1.0 \times 10^4$ to 1.5×10^4 on the sub-critical plateau of Figure 9 (Appendix A.5) with $C_{D,s} \approx 1.2$, so its contribution is removed as

$$D_{cyl} = D_{meas} - C_{D,s} \frac{1}{2} \rho U_\infty^2 d_s \ell_s, \quad (3.6)$$

after which the coefficients follow on the cylinder frontal area $D\ell = 5.5 \times 10^{-3} \text{ m}^2$. The correction removes 0.046 N and 0.103 N from the measured 0.20 N and 0.39 N, or 23 % and 26 %, and Table 2 compares both methods.

Table 2: Cylinder force coefficients from pressure integration and from the corrected transducer.

U_∞ (m/s)	Re	C_d		C_l	
		Pressure	Transducer	Pressure	Transducer
6	2.0×10^4	2.48	1.28	0.06	1.67
9	3.0×10^4	1.29	1.06	-0.05	0.96

3.2.4 Comparison of the Two Methods

The two methods of Table 2 agree to within 22 % in drag at $Re = 3.0 \times 10^4$ but differ by a factor of 1.9 at 2.0×10^4 . Integrating surface pressure captures only the form drag and omits skin friction, but Achenbach [7] measured that to be a few per cent of the total, so it cannot explain the gap. The base region can. Only seven taps (30° apart, 15° for taps 12 and 13) cover the separated face, where $\cos \theta \approx -1$, so those few readings dominate Equation (3.4). The 6 m/s base pressure of -3.1 is far deeper than the -0.8 at 9 m/s, and that alone doubles its C_d to 2.48. The strut coefficient is read off Figure 9 for an isolated cylinder, so a 20 % error in it shifts C_d by 6 to 7 %. End effects enter only the transducer [8]. A non-rotating cylinder carries no circulation and so no mean lift, which the integrated $C_l = +0.06$ and -0.05 confirm, yet the transducer reports 1.67 and 0.96. The lift

moves only from 0.20 N to 0.26 N, and the moment likewise, while q rises by 2.25 and the drag on the same instrument goes from 0.20 N to 0.39 N, so lift and moment carry an offset that the zeroing did not remove. Fitting $L = L_0 + C_l q D \ell$ gives $L_0 = 0.15$ N and $C_l = 0.40$, an exact fit on two points and so only an estimate, but one whose residual sits within the 0.4 to 0.6 r.m.s. side force of vortex shedding [9], [10], [11]. Conducting runs at three or more speeds would overdetermine the fit and show whether L_0 is the same at every speed.

3.2.5 Flow Regimes with Reynolds Number

The wake changes with Reynolds number through the boundary-layer state at separation, as Figure 3 shows, and Table 4 (Appendix A.6) details the full sequence. Sub-critically, for $300 < Re < 2 \times 10^5$, the boundary layer remains laminar up to separation, carries little near-wall momentum and so detaches early, near 80° [4]: the cylinder trails a wide wake at almost uniform base pressure, sheds a Kármán street [5], [9], and C_d plateaus near 1.2 [1]. Through the critical range, $2 \times 10^5 < Re < 5 \times 10^5$, the layer undergoes laminar-to-turbulent transition ahead of separation, so turbulent mixing drags high-momentum fluid to the wall, the flow holds to about 140° , the wake collapses and C_d falls to 0.3 [7], [12]. This is the drag crisis, where going faster lowers C_d . Post-critically, beyond $Re \approx 5 \times 10^5$, transition creeps onto the front face, separation follows back to near 120° and C_d recovers to 0.5 to 0.7 [11]. Both runs sit well below this range, in the sub-critical regime: separation is inferred near 80° in Figure 2, the base pressure is flat, and $C_d = 1.29$ at 9 m/s matches the plateau of Figure 9. The drag crisis is unreachable using this wind tunnel and a 50 mm cylinder, as this requires a U_∞ of 60 to 151 m/s.

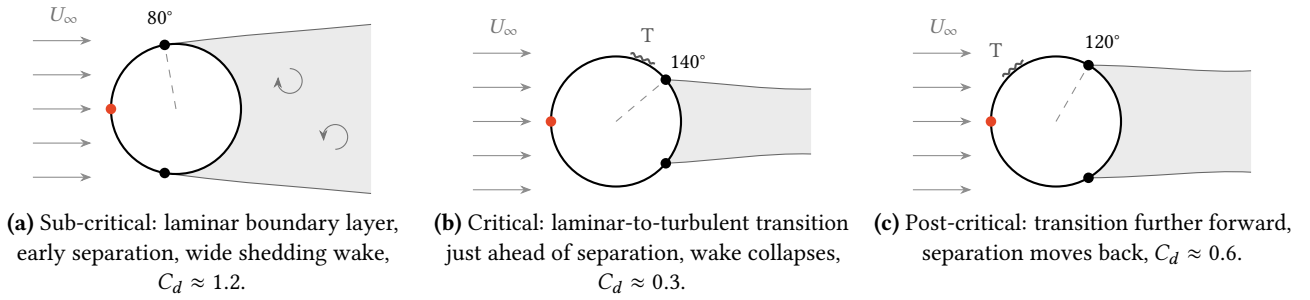


Figure 3: Cylinder wake from sub-critical to post-critical. Red marks the forward stagnation point, black is separation at the labelled θ , T the laminar-to-turbulent transition, and shading is the wake.

3.3 NACA 23012 Aerofoil

3.3.1 Surface Pressure Distributions

The section was tested at a nominal $U_\infty = 12$ m/s, to which the coefficients are referenced, giving $q = 87.4$ Pa and $Re = 1.4 \times 10^5$ on the 175 mm chord. Each of the twenty-six taps (Appendix A.9) then gives $C_p = (p - p_\infty)/q$ from its time-averaged reading. Figure 4 plots every angle of attack on one axis, suction plotting upward, with Figure 10 decomposing this into a grid of subplots for each α case. At $\alpha = 0^\circ$, the tufts lay flat and the mild suction of attached flow shows the camber carrying lift at zero incidence. From 2° to 16° , the upper-surface suction peak deepens to $C_p \approx -4.7$ while the pressure recovers smoothly to the trailing edge, so the flow stays attached, though the tufts showed rear-surface unsteadiness from 6° and strongly turbulent aft flow by 15° . At 18° and 20° , the peak collapses onto a nearly uniform plateau of $C_p \approx -1.2$ across the whole upper surface, the signature of full separation, which the tufts confirmed visually at 20° . The error bars show the same change. With the flow attached, the r.m.s. fluctuation sits near 0.007 at every tap. By 10° , it has grown to 0.08 at the suction peak, and at 18° the smallest value anywhere on the upper surface is 0.069, so by then the whole surface is unsteady rather than just the leading edge. Negative angles mirror this trend, with the lower surface now carrying the suction. At $\alpha = -10^\circ$, its pressure flattens aft of mid-chord while the upper surface stays attached.

3.3.2 Force and Moment Coefficients

With the pressure acting normal to a small panel at each tap, the normal and axial force coefficients are

$$C_n = \frac{1}{c} \int_0^c (C_{p,l} - C_{p,u}) dx \quad (3.7)$$

$$C_a = \frac{1}{c} \int_0^c \left(C_{p,u} \frac{dy_u}{dx} - C_{p,l} \frac{dy_l}{dx} \right) dx \quad (3.8)$$

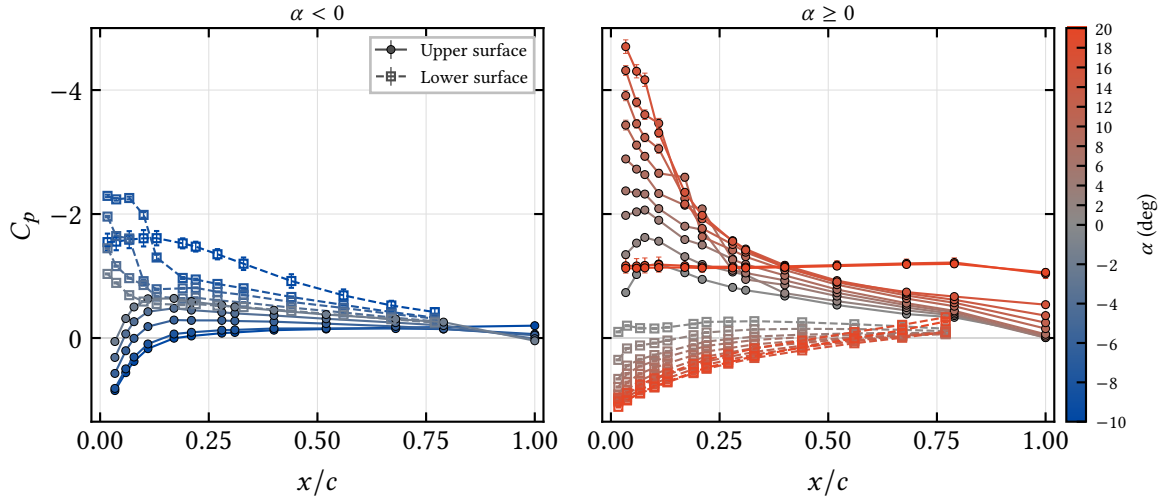


Figure 4: Surface pressure coefficient on the NACA 23012 at $Re = 1.4 \times 10^5$, suction plotting upward. Error bars are the r.m.s. fluctuation of C_p over the 2500 recorded frames (Appendix A.8).

and the moment about the quarter-chord point is

$$C_{m,c/4} = \frac{1}{c^2} \left[\int_0^c (C_{p,u} - C_{p,l}) \left(\frac{c}{4} - x \right) dx + \int_0^c \left(C_{p,u} \frac{dy_u}{dx} y_u - C_{p,l} \frac{dy_l}{dx} y_l \right) dx \right], \quad (3.9)$$

with ordinates and slopes from Appendix A.10. Each integral is evaluated by the composite trapezoidal rule, with each surface integrated over its own tap stations (Appendix A.9) because the upper and lower sets sit at different x/c . Rotating by the angle of attack gives

$$C_l = C_n \cos \alpha - C_a \sin \alpha \quad (3.10)$$

$$C_d = C_n \sin \alpha + C_a \cos \alpha \quad (3.11)$$

Figure 5 shows a lift curve linear between -8° and 10° with a fitted slope of $6.0/\text{rad}$, within 4% of the thin-aerofoil value of 2π per radian, and a zero crossing at $\alpha_0 \approx -2.0^\circ$ against the published -1.2° [13]. Lift peaks at $C_l = 1.51$ at 16° and drops to 1.16 at 18° , bracketing the stall, slightly above the observation of stall at about 14° from the tuft visualisation. The quarter-chord moment holds between $+0.013$ and $+0.097$ below the stall, as expected of the aerodynamic centre, then jumps to $+0.259$ at 18° when the suction peak collapses. Its positive sign against the small negative published value [13] is likely attributed to the missing lower-surface taps aft of $x/c = 0.77$, which the moment arm weights heavily. Appendix B compares the whole set against an XFOIL simulation at the same Reynolds number, which reproduces the linear slope but stalls 2° earlier at a 10% lower peak.

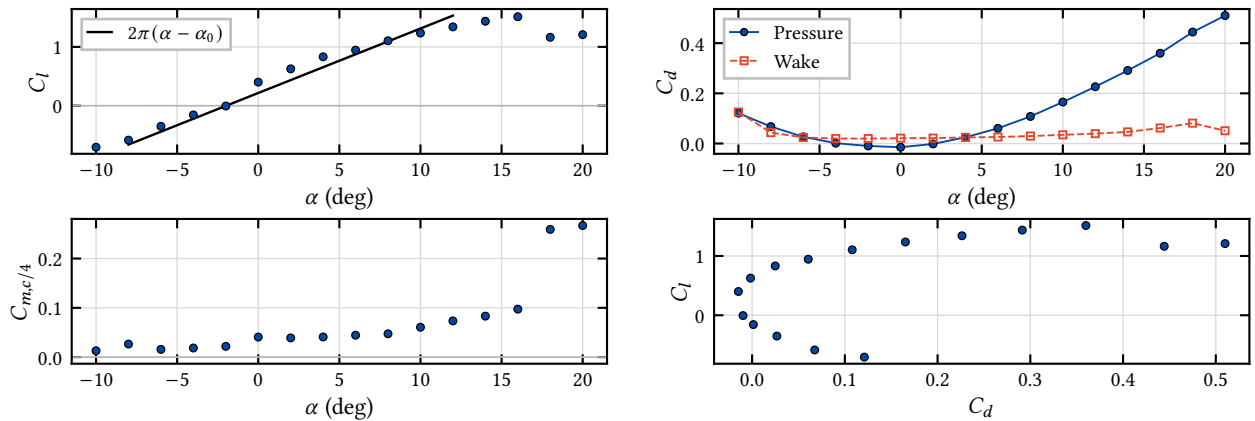


Figure 5: Coefficients and drag polar at $Re = 1.4 \times 10^5$ from pressure integration, with the wake drag of Section 3.3.4 on the C_d panel.

3.3.3 Wake Velocity Profiles

The rake carries fifteen total-pressure probes at 8 mm pitch. The static pressure has recovered to p_∞ by the measuring station, so Bernoulli's equation gives the local velocity at each probe as

$$\frac{u(y)}{U_\infty} = \sqrt{\frac{p_0(y) - p_\infty}{q}}, \quad (3.12)$$

with $p_0(y)$ the measured total pressure and p_∞ the reading of the probes lying clear of the wake, less q . Near zero incidence, Figure 6 shows a narrow velocity deficit, spanning about two probe spacings and reaching a minimum of $u/U_\infty = 0.87$, formed by the two surface boundary layers merging at the trailing edge. The deficit then deepens and widens steadily with angle, its centre migrating from $y/c = 0.05$ at 0° to 0.18 at 16° as the stream is deflected, and its minimum falling to 0.64 at 14° with the flow still attached. Past the stall it changes shape entirely: at 18° the eight central probes read a total pressure at or below p_∞ , so the velocity there is effectively zero, and at 20° ten probes do. In both cases, the outermost probe on the negative- y side has not recovered to U_∞ , so the wake is wider than the rake's 112 mm span and part of it is missed. The same broadening appears at -10° , minimum 0.59 , where the lower surface has separated instead.

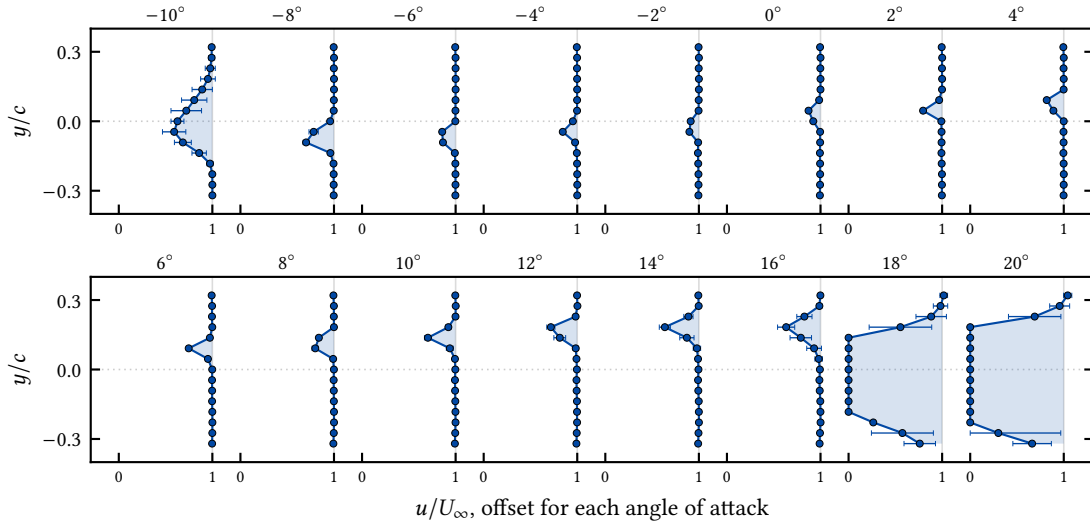


Figure 6: Normalised wake profiles at each angle of attack, every trace offset and read vertically against its own gridline at $u/U_\infty = 1$. The shaded area is the velocity deficit, y is positive towards port 47, and the error bars are the r.m.s. fluctuation carried through Equation (3.12) (Appendix A.8).

3.3.4 Drag from the Wake Integral

The momentum deficit gives the profile drag through the wake integral of Jones [2], evaluated by the trapezoidal rule over the fifteen ports,

$$C_d = \frac{2}{c} \int_{-\infty}^{\infty} \frac{u}{U_\infty} \left(1 - \frac{u}{U_\infty}\right) dy. \quad (3.13)$$

In the C_d panel of Figure 5, the wake integral gives 0.021 near zero incidence, against the 0.012 XFOIL predicts at the same Reynolds number, while the pressure integration returns small negative values, since it misses skin friction and at low angles is a small difference of large truncated-tap contributions, so the wake value is the reliable one. The two agree at $C_d = 0.12$ at -10° , where separated form drag dominates, but beyond 10° , the pressure value grows far faster, partly through the $C_n \sin \alpha$ rotation that enters only the pressure result. Past the stall, the wake values of 0.081 and 0.051 are underestimates, since Figure 6 shows it spilling past the rake, making the 18° to 20° drop an artefact of limits that no longer enclose the whole deficit.

4 Conclusion

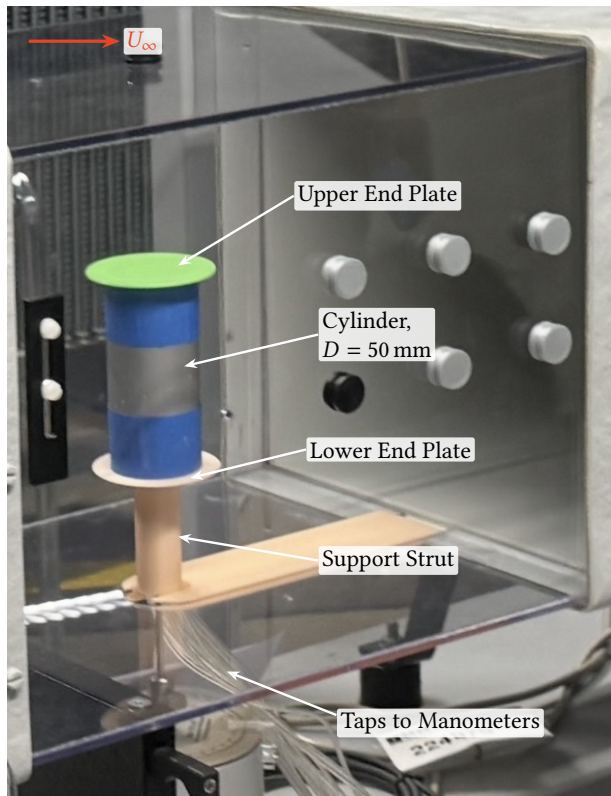
Using the air density of 1.214 kg/m^3 and viscosity of $1.828 \times 10^{-5} \text{ kg/(m s)}$ measured for Task 0, the cylinder distributions followed potential flow forward of laminar separation near 80° and flattened aft of it, placing both sub-critical runs on the drag plateau with $C_d = 1.29$ from pressure integration at $\text{Re} = 3.0 \times 10^4$ against 1.06 from the transducer. The aerofoil showed attached flow to 16° and full separation by 18° , a lift-curve slope of $6.0/\text{rad}$, a zero-lift angle of -2.0° , a maximum lift of 1.51 and a wake profile drag of 0.021 , with the two drag estimates agreeing where form drag dominates, meeting every objective of Section 1.

References

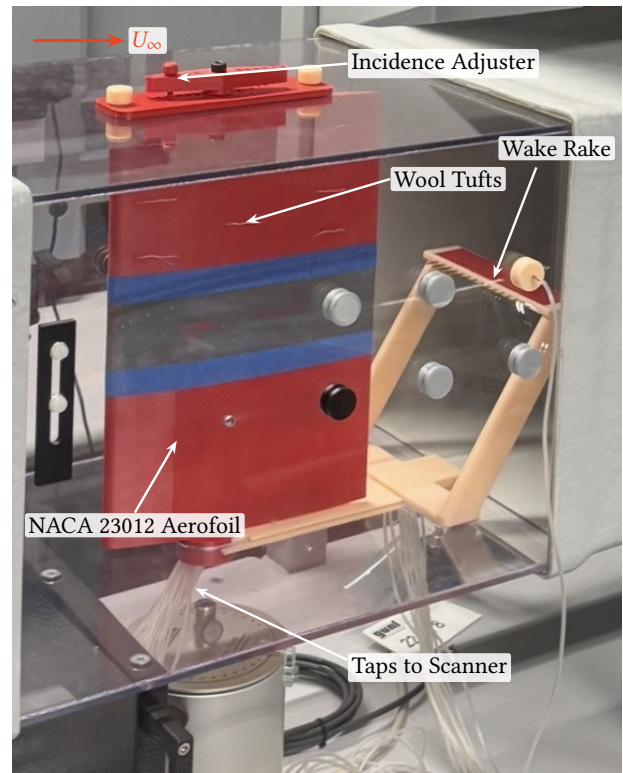
- [1] J. D. Anderson, *Fundamentals of Aerodynamics*, 6th ed. New York, NY, USA: McGraw-Hill Education, 2017.
- [2] B. M. Jones, *The Measurement of Profile Drag by the Pitot-Traverse Method* (Aeronautical Research Committee Reports and Memoranda 1688). London, UK: H.M. Stationery Office, 1936.
- [3] W. Sutherland, 'Lii. the viscosity of gases and molecular force,' *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, vol. 36, no. 223, pp. 507–531, 1893.
- [4] H. Jiang, 'Separation angle for flow past a circular cylinder in the subcritical regime,' *Physics of Fluids*, vol. 32, no. 1, p. 014 106, 2020.
- [5] A. Roshko, 'On the development of turbulent wakes from vortex streets,' California Institute of Technology, Tech. Rep. NACA Report 1191, 1954. [Online]. Available: <https://ntrs.nasa.gov/citations/19930092207>
- [6] R. L. Panton, *Incompressible Flow*. New York, NY, USA: Wiley-Interscience, 1984.
- [7] E. Achenbach, 'Distribution of local pressure and skin friction around a circular cylinder in cross-flow up to $Re = 5 \times 10^6$,' *Journal of Fluid Mechanics*, vol. 34, no. 4, pp. 625–639, 1968.
- [8] J. B. Barlow, W. H. Rae and A. Pope, *Low-Speed Wind Tunnel Testing*, 3rd ed. New York, NY, USA: John Wiley & Sons, 1999.
- [9] E. Achenbach and E. Heinecke, 'On vortex shedding from smooth and rough cylinders in the range of reynolds numbers 6×10^3 to 5×10^6 ,' *Journal of Fluid Mechanics*, vol. 109, pp. 239–251, 1981.
- [10] C. Norberg, 'Fluctuating lift on a circular cylinder: Review and new measurements,' *Journal of Fluids and Structures*, vol. 17, no. 1, pp. 57–96, 2003.
- [11] G. Schewe, 'On the force fluctuations acting on a circular cylinder in crossflow from subcritical up to transcritical reynolds numbers,' *Journal of Fluid Mechanics*, vol. 133, pp. 265–285, 1983.
- [12] H. Schlichting and K. Gersten, *Boundary-Layer Theory*, 9th ed. Berlin, Germany: Springer, 2017.
- [13] I. H. Abbott and A. E. von Doenhoff, *Theory of Wing Sections*. New York, NY, USA: Dover Publications, 1959.
- [14] J. H. Lienhard, 'Synopsis of lift, drag and vortex frequency data for rigid circular cylinders,' Washington State University, College of Engineering, Pullman, WA, USA, Research Division Bulletin 300, 1966.
- [15] B. M. Sumer and J. Fredsøe, *Hydrodynamics Around Cylindrical Structures*, Revised. Singapore: World Scientific, 2006.
- [16] M. Drela and H. Youngren, *XFOIL 6.9 user primer*, MIT Department of Aeronautics, Astronautics and Aircraft, Inc., Nov. 2001. [Online]. Available: https://web.mit.edu/aeroutil_v1.0/xfoil_doc.txt

Appendix A Raw Data

A.1 Test Section Photographs



(a) Cylinder rig, looking downstream.



(b) Aerofoil rig with the wake rake behind it.

Figure 7: The test section as set up, annotated. Both models span the tunnel vertically, the cylinder between end plates and the aerofoil between the tunnel walls, with the pressure tubing leaving through the floor.

A.2 Cylinder Pressure Tap Mapping

The taps alternate sides of the cylinder, as shown in Figure 8. Table 3 lists the signed angular location of each tube, measured from the front stagnation point with angles above the axis taken as positive. The analysis of Section 3.2.1 plots against the magnitude of θ , which takes the mean flow as symmetric about the freestream direction.

Table 3: Manometer tube number against angular position on the cylinder.

Tube no.	1	2	3	4	5	6	7	8	9	10	11	12	13
θ (°)	0	15	-30	45	-60	75	-90	105	-120	135	-150	165	180

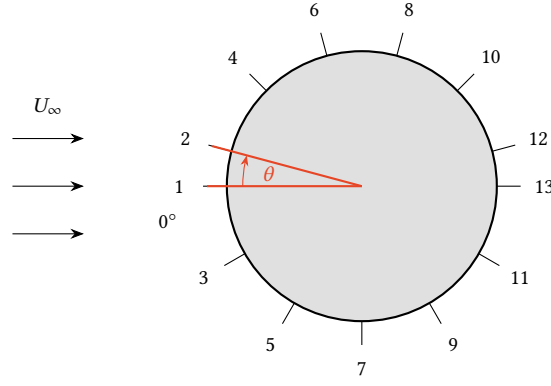


Figure 8: Orientation of the pressure taps on the cylinder. Odd-numbered taps past port 1 sit below the horizontal axis and even-numbered taps above it, with θ measured from the front stagnation point.

A.3 Cylinder Force Coefficients from the Tap Pressures

The brief quotes the cylinder coefficients directly, below its figure 2, in the form evaluated in Section 3.2.2. They follow from its equation 3, which assumes the pressure acts normal to a panel at each tap and resolves it into a component normal to the chord and one along it,

$$C_n = \frac{1}{c} \int_0^c (C_{p,l} - C_{p,u}) dx, \quad C_a = \frac{1}{c} \int_0^c \left(C_{p,u} \frac{dy_u}{dx} - C_{p,l} \frac{dy_l}{dx} \right) dx, \quad (\text{A.1})$$

after which equation 5 of the brief rotates them through the angle of attack. A cylinder has no chord line to incline, so $\alpha = 0$ and the rotation is the identity, so $C_l = C_n$ and $C_d = C_a$ directly, with the chord replaced by the diameter, $c = D$.

Taking θ from the front stagnation point and positive above the axis, the surface of a circle of diameter D is

$$x = \frac{D}{2} (1 - \cos \theta), \quad y_{u,l} = \pm \frac{D}{2} \sin \theta, \quad \text{so} \quad dx = \frac{D}{2} \sin \theta d\theta, \quad \frac{dy_{u,l}}{dx} = \pm \cot \theta. \quad (\text{A.2})$$

Substituting Equation (A.2) into Equation (A.1), the $\sin \theta$ of the arc element cancels the $\cot \theta$ of the surface slope in C_a and supplies the $\sin \theta$ weighting in C_n , giving

$$\begin{aligned} C_d &= \frac{1}{2} \int_0^\pi (C_{p,u} + C_{p,l}) \cos \theta d\theta = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta, \\ C_l &= \frac{1}{2} \int_0^\pi (C_{p,l} - C_{p,u}) \sin \theta d\theta = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta, \end{aligned} \quad (\text{A.3})$$

which is the form quoted in the brief. The same result follows in one step from the pressure force $d\mathbf{F} = -p \hat{\mathbf{n}} \frac{D}{2} d\theta$ on the arc element, with outward normal $\hat{\mathbf{n}} = (-\cos \theta, \sin \theta)$ in freestream-aligned axes, where the factor $\frac{1}{2}$ is the arc radius $\frac{D}{2}$ referred to the frontal area D per unit span.

Two properties of Equation (A.3) are used in Section 3.2.2. First, the freestream static pressure drops out, because $\int_0^{2\pi} \cos \theta d\theta = \int_0^{2\pi} \sin \theta d\theta = 0$, so a uniform error in C_p from a misread reference leaves both coefficients unchanged. On the discrete contour of Table 3 the same sums evaluate to -2×10^{-17} and -6×10^{-4} . Second, substituting the potential-flow distribution $C_p = 1 - 4 \sin^2 \theta$ returns $C_d = C_l = 0$, which is d'Alembert's paradox and confirms the sign convention. That convention rests on the origin of θ . Measured instead from the downstream axis, the front stagnation point moves to 180° , $\cos \theta$ changes sign and both integrals carry $-\frac{1}{2}$. The two forms return identical coefficients, but pairing the front-referenced θ of Table 3 with $-\frac{1}{2}$ in C_d would report thrust on a bluff body.

The thirteen taps of Table 3 alternate sides, so their signed angles already span the circle and close the contour, and the composite trapezoidal rule is applied over them directly. No symmetry is imposed, so C_l follows from the readings rather than from an assumption.

A.4 Numerical Evaluation of the Cylinder Integrals

Taking the taps in order of increasing θ and folding the negative angles of Table 3 onto $[0, 2\pi)$ gives the thirteen quadrature stations

$$\theta_i = 0^\circ, 15^\circ, 45^\circ, 75^\circ, 105^\circ, 135^\circ, 165^\circ, 180^\circ, 210^\circ, 240^\circ, 270^\circ, 300^\circ, 330^\circ, \quad (\text{A.4})$$

which are not uniformly spaced, the steps being 30° throughout except either side of the axis, where taps 1–2 and 12–13 give 15° . The contour is closed by carrying the first tap round a full turn,

$$\theta_{14} = \theta_1 + 2\pi, \quad C_{p,14} = C_{p,1}, \quad (\text{A.5})$$

so that the thirteen intervals $\Delta\theta_i = \theta_{i+1} - \theta_i$ span the whole circle, $\sum_{i=1}^{13} \Delta\theta_i = 2\pi$.

For any integrand $f(\theta)$ the composite trapezoidal rule on this non-uniform grid is the sum of the thirteen trapezia,

$$\int_0^{2\pi} f \, d\theta \approx \sum_{i=1}^{13} \frac{f_i + f_{i+1}}{2} \Delta\theta_i. \quad (\text{A.6})$$

Substituting $f = C_p \cos \theta$ into Equation (A.3) and carrying the $\frac{1}{2}$ of that equation through the $\frac{1}{2}$ of Equation (A.6) gives

$$C_d = \frac{1}{2} \int_0^{2\pi} C_p \cos \theta \, d\theta \approx \frac{1}{2} \sum_{i=1}^{13} \frac{C_{p,i} \cos \theta_i + C_{p,i+1} \cos \theta_{i+1}}{2} \Delta\theta_i, \quad (\text{A.7})$$

whose two halves combine into the quarter of Equation (3.4). The same substitution with $f = C_p \sin \theta$, carrying the leading minus sign, gives Equation (3.5). That quarter is therefore two separate halves, one from the arc element $\frac{D}{2} d\theta$ referred to the frontal area D and one from averaging the two end values of each trapezium. Evaluated on the readings of Table 5 they return $C_d = 2.477$ and $C_l = +0.059$ at 6 m/s, and 1.292 and -0.053 at 9 m/s.

Imposing instead the symmetry used for plotting in Section 3.2.1, by reflecting each reading so that $C_p(-\theta) = C_p(\theta)$, closes the contour on 24 uniform intervals of 15° and returns $C_d = 2.44$ and 1.26, within 1.5 % and 2.4 % of the signed values. The drag is therefore insensitive to which treatment is used, but the reflected distribution is even in θ , so $C_p \sin \theta$ is odd and C_l vanishes identically whatever the manometer recorded. The symmetric treatment cannot return a lift coefficient at all, which is why Section 3.2.2 integrates the signed angles. The small C_l it returns measures the departure from that assumed symmetry.

A.5 Cylinder Drag Coefficient against Reynolds Number

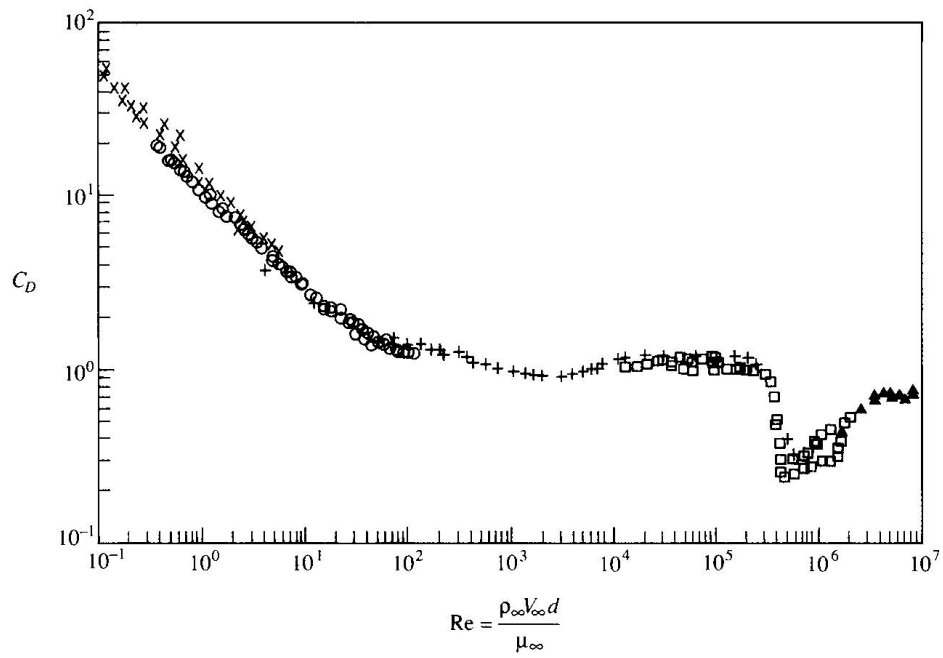
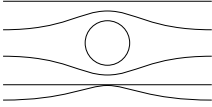
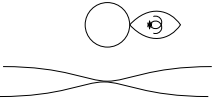
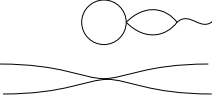
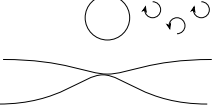
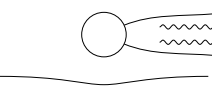
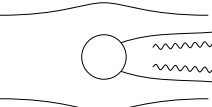


Figure 9: Variation of the drag coefficient of a circular cylinder with Reynolds number, reproduced from figure 1 of the laboratory brief. Data compiled by Panton [6], as reproduced by Anderson [1].

A.6 Flow Regimes of a Smooth Circular Cylinder

Table 4: Regimes of flow about a smooth circular cylinder in steady current, after Lienhard [14] as reproduced by Sumer and Fredsøe [15]. The boundaries are the ones usually quoted and vary between sources and with surface roughness. Figure 3 expands the last three rows, which span the sub-critical to post-critical range of Section 3.2.5.

Reynolds number	Regime	Flow form	Characteristic
$Re \rightarrow 0$	Creeping flow		Steady, attached, no wake
$4 < Re < 40$	Fixed vortex pair		Steady, symmetric separation
$40 < Re < 90$	Onset of shedding		Laminar, unstable wake
$90 < Re < 300$	Kármán vortex street		Fully periodic laminar shedding
$300 < Re < 2 \times 10^5$	Sub-critical		Laminar separation, turbulent vortex street
$2 \times 10^5 < Re < 5 \times 10^5$	Critical		Laminar separation, turbulent reattachment, turbulent separation
$Re > 5 \times 10^5$	Post-critical		Fully turbulent separation

A.7 Cylinder Manometer and Force Readings

Table 5 lists the manometer column heights recorded at both tunnel speeds, and Table 6 the force transducer readings. The transducer values include the contribution of the support structure, which is subtracted in Section 3.2.3.

Table 5: Raw manometer column heights in mm of water. Tube 14 is connected to the tunnel static port.

Tube no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$U_\infty = 6 \text{ m/s}$	386	386	388	390	393	394	393	394	394	394	394	394	395	388
$U_\infty = 9 \text{ m/s}$	386	387	391	393	397	397	396	396	397	398	397	397	398	394

Table 6: Force transducer readings for the cylinder and support structure.

$U_\infty \text{ (m/s)}$	Lift (N)	Drag (N)	Moment (N m)
6	0.20	0.20	-0.03
9	0.26	0.39	-0.04

A.8 Error Bars

The manometer gives a single reading per tap, so the bars in Figure 2 are the reading resolution of the instrument rather than a statistical interval. The HM 170.50 scale is graduated in millimetres of water and was read upright to the nearest half graduation, so $\delta h = 0.5$ mm and Equation (3.3) gives

$$\delta C_p = \frac{\rho_w g \delta h}{\frac{1}{2} \rho U_\infty^2} = \frac{(1000)(9.81)(5 \times 10^{-4})}{q} = \frac{4.91 \text{ Pa}}{q} = \begin{cases} 0.22, & q = 21.8 \text{ Pa (6 m/s)} \\ 0.10, & q = 49.2 \text{ Pa (9 m/s)} \end{cases} \quad (\text{A.8})$$

The reference reading on tube 14 carries the same resolution, but being common to every tap it shifts the whole distribution rather than scattering individual points, so it is excluded from the bars. The Scanivalve instead records 2500 frames per port, and the bars in Figure 4 are the r.m.s. fluctuations of those frames divided by q . The wake bars of Figure 6 are that same quantity carried through Equation (3.12),

$$\delta \left(\frac{u}{U_\infty} \right) = \frac{s/q}{2u/U_\infty}, \quad (\text{A.9})$$

which is suppressed below $u/U_\infty = 0.3$, where the square root makes the conversion ill-conditioned, and truncated at zero on the low side because the velocity cannot be negative.

A.9 Aerofoil Pressure Tap Locations

Table 7: Chordwise positions of the NACA 23012 pressure taps, after tables 2 and 3 of the brief. Tubes 1 to 13 serve the lower surface and 14 to 26 the upper.

Tube no.	1	2	3	4	5	6	7	8	9	10	11	12	13
x/c (lower)	0.017	0.037	0.067	0.10	0.13	0.19	0.22	0.27	0.33	0.44	0.56	0.67	0.77
Tube no.	14	15	16	17	18	19	20	21	22	23	24	25	26
x/c (upper)	0.034	0.059	0.078	0.11	0.17	0.21	0.28	0.31	0.40	0.52	0.68	0.79	1.00

A.10 NACA 23012 Section Definition

The surface ordinates $y_{u,l}$ and slopes $\frac{dy_{u,l}}{dx}$ that Equations (3.7)–(3.9) require are taken from the NACA five-digit definition in appendix 2 of the brief. For the 230 camber line the parameters are $m = 0.2025$ and $k_1 = 15.957$, and the thickness is $t = 0.12$. With x in chords, the camber line and its slope are

$$y_c = \begin{cases} \frac{k_1}{6} (x^3 - 3mx^2 + m^2(3-m)x), & x \leq m, \\ \frac{k_1}{6} m^3 (1-x), & x > m, \end{cases} \quad (\text{A.10})$$

and the thickness distribution is

$$y_t = \frac{t}{0.2} \left(0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4 \right). \quad (\text{A.11})$$

The surfaces then follow as $y_u = y_c + y_t$ and $y_l = y_c - y_t$, with the slopes differentiated analytically rather than from the tap spacing, so that the axial and moment integrands are exact at each station.

The position of maximum camber, $p = 0.15$ for the 230 line, does not appear in Equation (A.10). It is the design quantity that fixes m , through $p = m \left(1 - \sqrt{m/3} \right)$, and only m and k_1 enter the polynomial. Substituting $m = 0.2025$ returns $p = 0.1499$, and the camber line does peak there, at $y_c = 0.0184$.

A.11 Aerofoil Pressure Distributions by Angle of Attack

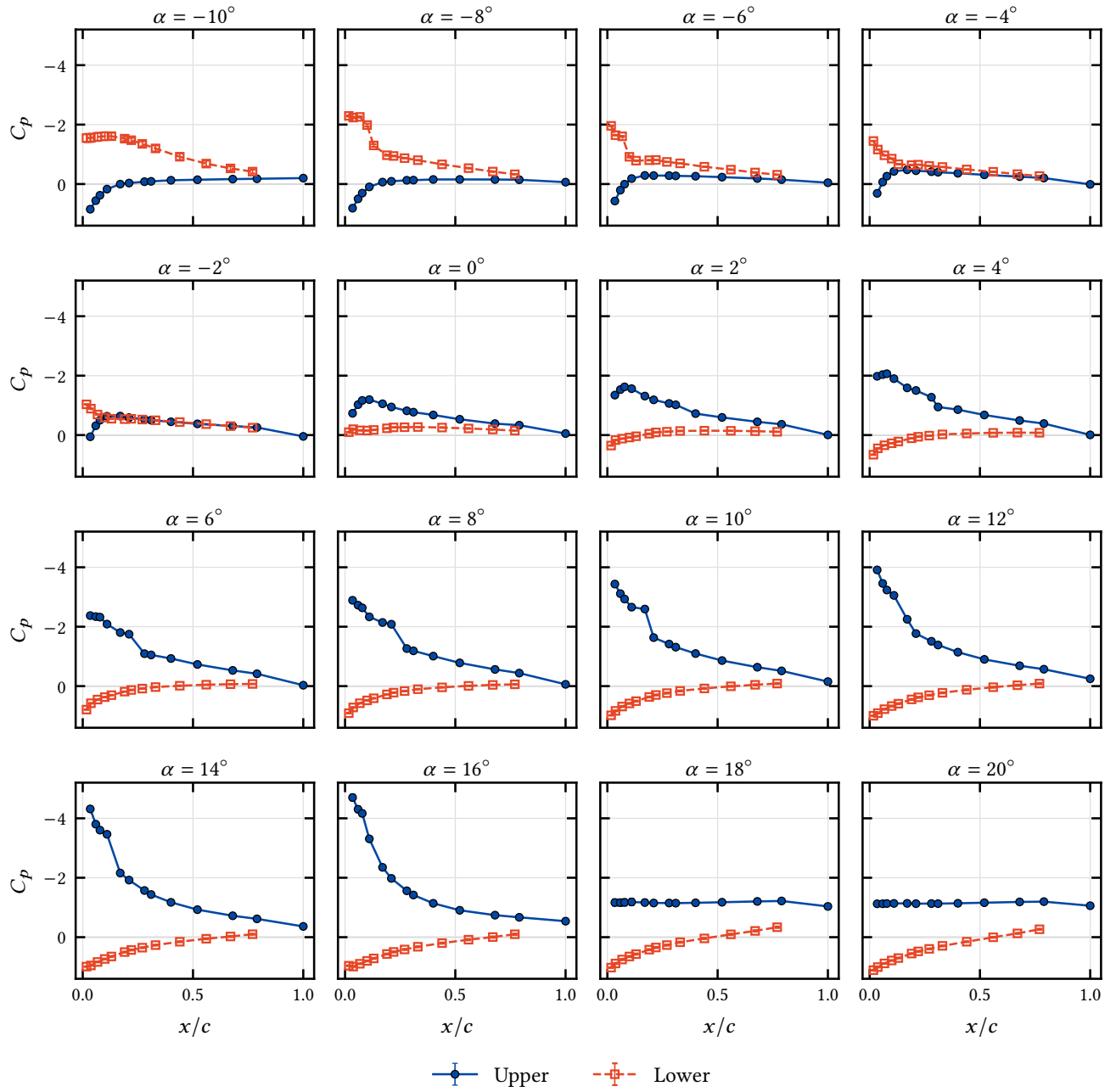


Figure 10: The data of Figure 4 decomposed by angle of attack, with the same r.m.s. bars and the vertical axis inverted so suction plots upward.

A.12 Aerofoil Surface and Wake Pressures

Tables 8 and 9 list the time-averaged Scanivalve readings for every used channel at each angle of attack, in pascals gauge relative to the scanner's atmospheric reference. Each value is the mean of the 2500 recorded frames, and the full time series are available in the archived data files listed with the analysis code of Appendix C.

Table 8: Mean surface tap pressures (Pa) at each angle of attack ($^{\circ}$). Ports 1 to 13 are the lower surface and 14 to 26 the upper surface.

Port	-10	-8	-6	-4	-2	0	2	4	6	8	10	12	14	16	18	20
1	-233	-299	-269	-225	-188	-107	-67	-41	-29	-19	-12	-11	-11	-14	-12	-10
2	-234	-294	-242	-200	-176	-115	-84	-60	-48	-36	-26	-19	-15	-12	-24	-19
3	-237	-296	-239	-183	-159	-112	-88	-69	-59	-48	-38	-31	-25	-20	-36	-30
4	-239	-272	-178	-173	-150	-111	-91	-75	-66	-56	-47	-40	-34	-28	-45	-38
5	-239	-212	-167	-157	-146	-113	-95	-80	-72	-63	-54	-47	-41	-36	-52	-46
6	-232	-183	-168	-154	-145	-118	-102	-89	-82	-74	-66	-59	-54	-49	-65	-59
7	-227	-181	-169	-155	-146	-121	-106	-94	-87	-79	-72	-65	-60	-55	-71	-65
8	-216	-174	-163	-152	-144	-121	-108	-97	-91	-84	-78	-72	-67	-62	-78	-73
9	-203	-168	-159	-149	-142	-122	-110	-101	-96	-89	-84	-79	-75	-70	-87	-81
10	-179	-156	-149	-141	-136	-120	-111	-103	-100	-95	-91	-88	-85	-81	-98	-94
11	-158	-145	-141	-134	-130	-118	-111	-105	-103	-99	-98	-95	-93	-91	-110	-107
12	-144	-135	-132	-128	-125	-115	-110	-106	-104	-102	-102	-101	-100	-99	-120	-118
13	-135	-127	-126	-122	-120	-111	-108	-105	-105	-104	-106	-106	-107	-107	-131	-130
14	-24	-27	-48	-71	-93	-162	-216	-271	-306	-351	-398	-440	-475	-509	-203	-205
15	-49	-55	-80	-104	-126	-188	-232	-276	-303	-336	-370	-400	-430	-474	-203	-205
16	-65	-71	-98	-122	-142	-200	-240	-279	-301	-329	-354	-381	-413	-462	-204	-206
17	-83	-90	-114	-136	-153	-202	-235	-265	-281	-302	-330	-365	-401	-387	-205	-206
18	-98	-104	-124	-140	-154	-190	-213	-237	-256	-285	-325	-295	-287	-304	-203	-206
19	-101	-106	-123	-138	-149	-181	-202	-230	-251	-280	-241	-253	-266	-271	-202	-206
20	-105	-109	-123	-135	-144	-169	-191	-210	-194	-209	-222	-230	-235	-235	-202	-206
21	-106	-110	-122	-133	-142	-165	-187	-181	-190	-202	-213	-219	-223	-222	-202	-206
22	-109	-112	-121	-130	-137	-157	-161	-173	-180	-187	-194	-198	-200	-198	-203	-207
23	-111	-112	-119	-126	-131	-145	-150	-158	-162	-167	-173	-177	-179	-177	-204	-208
24	-113	-112	-115	-120	-125	-132	-137	-141	-145	-147	-154	-158	-161	-163	-207	-211
25	-114	-111	-111	-116	-121	-127	-130	-132	-135	-137	-143	-148	-152	-157	-208	-212
26	-116	-104	-102	-98	-94	-103	-99	-99	-101	-104	-112	-120	-130	-145	-192	-200

Table 9: Mean wake rake and tunnel static pressures (Pa) at each angle of attack (°). Ports 33 to 47 are the rake totals, port 48 the tunnel-inlet static and port 49 the side-wall static.

Port	-10	-8	-6	-4	-2	0	2	4	6	8	10	12	14	16	18	20
33	-11	-11	-11	-11	-12	-12	-11	-12	-11	-12	-12	-12	-12	-12	-51	-69
34	-11	-11	-11	-11	-11	-12	-11	-11	-11	-11	-12	-12	-11	-12	-73	-99
35	-11	-11	-11	-11	-12	-11	-11	-11	-11	-11	-11	-11	-10	-11	-95	-127
36	-15	-12	-11	-11	-12	-11	-12	-11	-11	-11	-11	-11	-11	-11	-121	-152
37	-34	-17	-12	-12	-13	-12	-12	-12	-12	-11	-11	-11	-11	-11	-141	-171
38	-57	-55	-33	-15	-12	-11	-12	-12	-11	-11	-12	-12	-12	-12	-158	-186
39	-68	-45	-34	-36	-28	-11	-12	-12	-12	-12	-12	-12	-12	-13	-166	-190
40	-64	-18	-11	-19	-25	-24	-12	-11	-11	-11	-11	-12	-11	-12	-171	-194
41	-53	-11	-11	-11	-11	-32	-43	-29	-19	-13	-12	-12	-12	-14	-172	-199
42	-42	-11	-11	-11	-11	-13	-16	-40	-50	-42	-21	-14	-14	-22	-162	-202
43	-29	-11	-11	-11	-11	-11	-11	-11	-16	-37	-55	-40	-31	-44	-129	-185
44	-19	-11	-11	-11	-11	-11	-11	-11	-11	-11	-24	-53	-63	-63	-75	-131
45	-15	-11	-11	-12	-11	-11	-11	-12	-12	-11	-11	-14	-29	-38	-33	-66
46	-12	-11	-11	-11	-11	-11	-11	-11	-12	-11	-11	-11	-11	-13	-17	-27
47	-12	-11	-12	-12	-11	-12	-12	-12	-12	-12	-11	-11	-12	-11	-11	-12
48	-123	-117	-113	-110	-108	-104	-102	-101	-101	-101	-104	-106	-109	-111	-139	-148
49	-104	-106	-106	-106	-106	-105	-104	-103	-102	-101	-100	-98	-97	-95	-91	-86

A.13 Manometer Datasheet

The cylinder surface pressures of Section 3.2 were read from the GUNT HM 170.50 sixteen-tube manometer described in Section 2. The specification page of its manufacturer datasheet is reproduced on the next page, and the specifications relevant to this report are the sixteen glass tubes, the millimetre water column scale, the selectable inclination of 1:1, 1:2, 1:5 or 1:10, the quoted resolution of 0.01 mbar at 1:10 inclination and the maximum differential pressure of 60 mbar.

HM 170.50

16 tube manometers, 600mm



The illustration shows HM 170.50 together with HM 170

Learning objectives/experiments

- measurement of differential pressures

Specification

- [1] unit for differential pressure measurement/display of differential pressure characteristics
- [2] accessory for the wind tunnel HM 170
- [3] simultaneous display of up to 16 differential pressures
- [4] individual zero point adjustment possible
- [5] manometer inclination up to max. 1/10

Technical data

Tube manometer

- manometer tube
 - ▶ number: 16
 - ▶ material: glas
 - ▶ Ø 9mm, outside
 - ▶ Ø 7mm, inside
- inclination: 1:1, 1:2, 1:5, 1:10
- resolution: 0,01 mbar at inclination: 1:10
- zero point adjustment

Measuring ranges

- differential pressure: max. 60mbar

LxWxH: 670x220x750mm

Weight: approx. 26kg

Description

- **manometer for displaying differential pressure characteristics**
- **up to 16 differential pressures can be measured simultaneously**

The HM 170.50 manometer is used to simultaneously display up to 16 measured differential pressures in millimetres water column. The manometer utilises the principle of communicating tubes. The zero point is adjusted using a vertically sliding compensation chamber as a function of the measuring task. The resolution can (with a simultaneous reduction of the measuring range) be increased by inclining the unit. The manometers are fixed to the frame of HM 170.

HM 170.50 is recommended as an accessory for experiments on pressure distribution on aerofoils HM 170.22, HM 170.26, HM 170.27, on pressure distribution on a cylinder HM 170.23 and for experiments on wake measurement HM 170.28.

Scope of delivery

- 1 tube manometers, completely assembled

A.14 Pressure Scanner Datasheet

The aerofoil surface and wake pressures of Section 3.3 were recorded by the Scanivalve MPS 4264 miniature pressure scanner described in Section 2, whose specification page is reproduced on the next page. The module used here carries the four-inch water full-scale range, that is 996 Pa, quoted to 0.20 % of full scale, so the instrument accuracy is ± 2.0 Pa. Against the $q = 87.4$ Pa of Section 3.3 that is a systematic bound of ± 0.023 on every C_p in Figure 4, larger than the r.m.s. fluctuation at small incidence and comparable to it near the stall. The datasheet also gives the scan rate as 850 Hz per channel, or 2 500 Hz in fast mode over sixteen channels; the rate actually selected was not logged with the runs, so the 2500 frames of each record correspond to at least 2.9 s of sampling.

Specifications

Inputs

Px	0.042" [1.067mm] OD (standard) 0.031" [.787mm] OD (optional)
Cal, Ref, CTL, Prg	0.063" [1.600mm] OD

Full Scale Ranges: 4 inH₂O, 8 inH₂O, 1psid, 5psid, 15psid, 50psid
[995.4Pa, 1990.7Pa, 6.89kPa, 34.5kPa, 103.4kPa, 344.7kPa]

Accuracy*:	4 inH ₂ O: 0.20%FS
	8 inH ₂ O: 0.15%FS
	1psid: 0.06%FS
	5psid: 0.06%FS
	15psid: 0.06%FS
	50psid: 0.06%FS

Overpressure Capability:

4 inH ₂ O:	25x
8 inH ₂ O:	15x
1psid:	15x
5psid:	10x
15psid:	5x
50psid:	2x

A/D Resolution: 24-bit

Media

Compatibility: Gases compatible with silicon, silicone, aluminum, and Buna-N

Ethernet Connection: 100baseT, MDIX auto-crossing

External Trigger: 5-15Vdc, 6.5mA

Data Output Rate:

(samples/channel/second)

TCP/IP Binary: 850Hz

"Fast Mode": 2500Hz

Power Requirements: For EPx valve:

9-36Vdc, 3.5W
18-26Vdc, 5.5W

Control Pressure Requirements:

CPx - 65psi min / 120psi max
NPx - 90psi min / 120psi max
EPx - N/A

Mating Connectors

Ethernet: TE Connectivity PLG 8P8C Mini2
Power: TE Connectivity PLG 8P8C Mini1

Weight:

CPx/NPx - 6.59oz [186.9g]
EPx - 12.88oz [365g]

Maximum

Reference Pressure: 50 psig (345kPa)

Environment Pressure: 0.5 to 100 psia (3.4 to 690kPa abs)

Operating

Temperature: 0° to 70°C
-50°C to 125°C with TCU

Storage Temperature: 0° to 80°C

Humidity: 5 to 95% RH, Non-Condensing

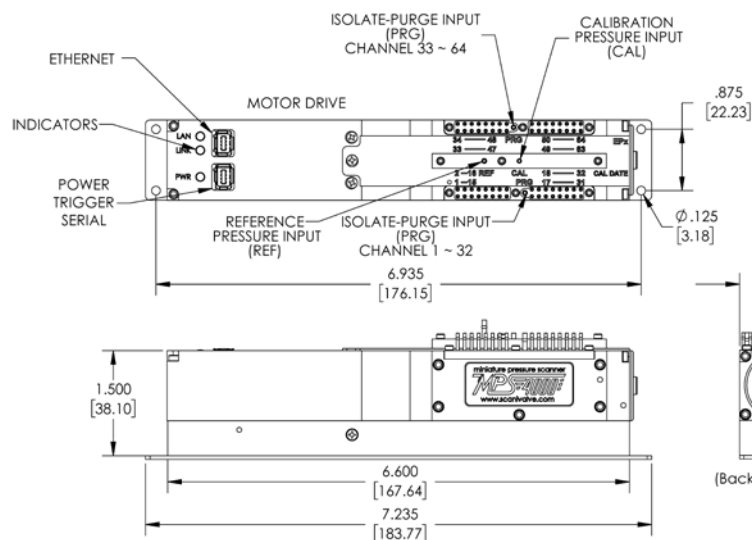
Standards:

MIL-STD-810G, Cat 24
CE
RoHSv3

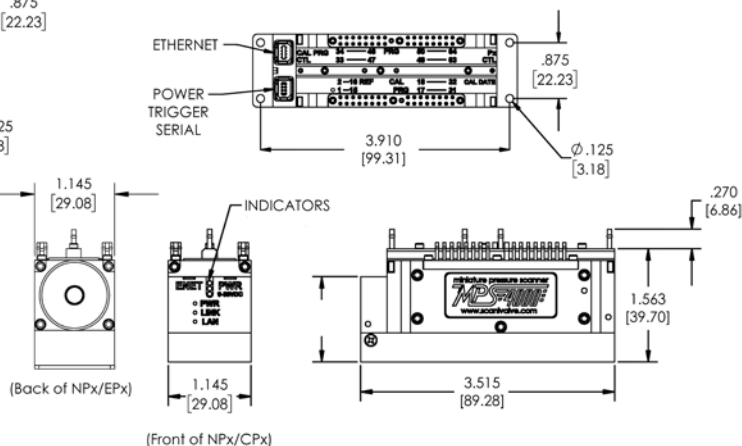
* Non-standard pressure ranges may result in a reduction of accuracy. Please contact Scanivalve for more information.

Dimensions Inches [mm]

Electric Valve (EPx)



Pneumatic Valve (NPx / CPx)



Scanivalve Headquarters

1722 N. Madson Street
Liberty Lake, WA 99019
Tel: 509-891-9970
800-935-5151
Fax: 509-891-9481
e-mail: scanco@scanivalve.com

Scanivalve

www.scanivalve.com

Appendix B Theoretical Validation with XFOIL

B.1 Computational Setup

Theoretical comparison curves were generated using XFOIL to evaluate the viscous aerodynamic response of the NACA 23012 profile at the experimental chord Reynolds number of $Re = 1.4 \times 10^5$. XFOIL couples an inviscid linear-vorticity panel method with a two-equation integral boundary-layer formulation. Rather than integrating surface pressures and skin friction directly, which is susceptible to panel discretisation error, XFOIL evaluates the boundary-layer momentum thickness θ_{te} and shape parameter H_{te} at the trailing edge and applies the Squire-Young relation:

$$C_d = 2 \theta_{te} \left(\frac{u_{te}}{U_\infty} \right)^{\frac{H_{te}+5}{2}} \quad (B.1)$$

This evaluates the total profile drag from the far-field wake momentum deficit, providing a direct theoretical counterpart to the experimental wake method [16].

B.2 Theoretical and Overlay Polars

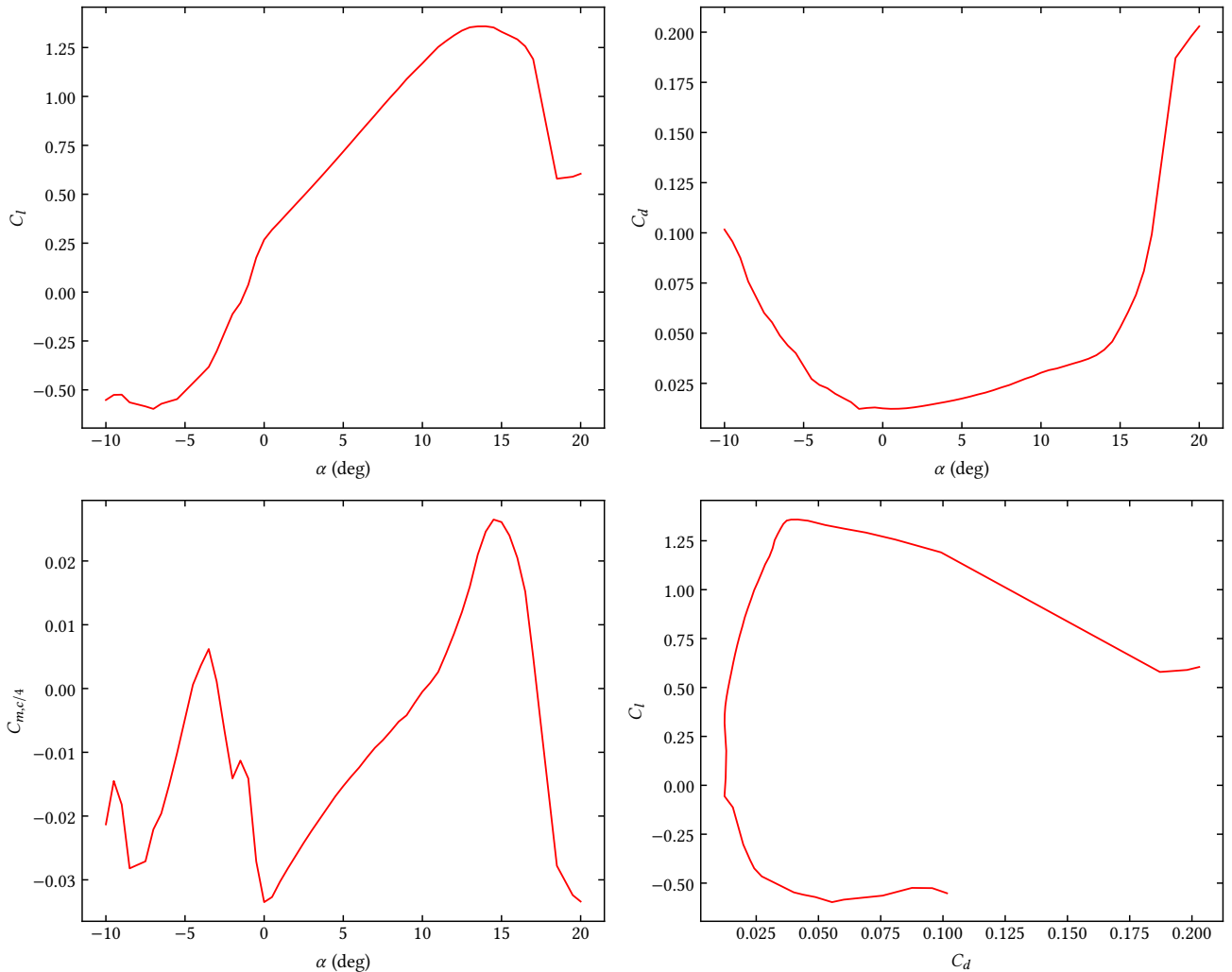


Figure 11: Theoretical aerodynamic coefficients of the NACA 23012 aerofoil from XFOIL at $Re = 1.4 \times 10^5$.

Figures 11 and 12 serve as useful validation of the experimentally obtained data. The theoretical lift data obtained from XFOIL validates the experimental pre-stall behaviour, predicting a linear lift slope approximately equal to that obtained experimentally. The experimental slope is slightly lower, likely due to tunnel wall boundary-layer growth and discrete tap resolution. Additionally, the theoretical zero-lift offset of $\alpha_0 = -1.20^\circ$ is in good agreement with the experimentally found value of $\alpha_0 = -2.0^\circ$, further validating the experimentally

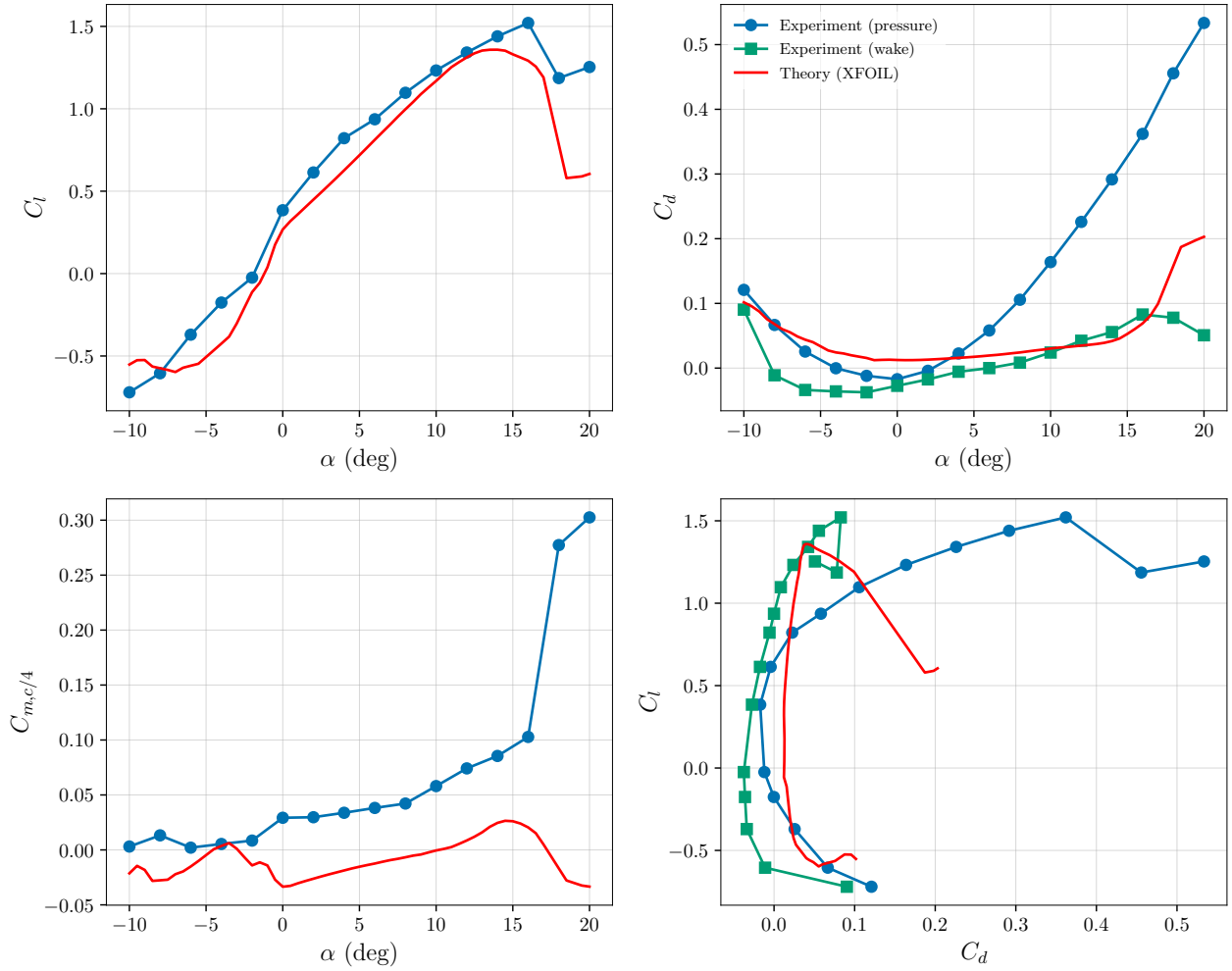


Figure 12: Comparison of experimental pressure-integrated data (blue) and wake data (green) against theoretical XFOIL data (red).

derived data. Both methods identify stall onset in the same region, with the experimental data identifying a slightly larger maximum lift coefficient ($C_{l,max,exp} = 1.51$ vs $C_{l,max,theo} = 1.36$) and slightly later stall onset ($\alpha_{stall,exp} = 16.0^\circ$ vs $\alpha_{stall,theo} = 14.0^\circ$), likely due to the discrete tap resolution.

Theoretical drag predictions from XFOIL agree closely with the experimental wake rake measurements across the entire attached flow regime ($-4^\circ \leq \alpha \leq 14^\circ$), with both methods capturing the characteristic low-drag bucket. XFOIL predicts a minimum profile drag of $C_{d,min} = 0.012$ at $\alpha = -1.5^\circ$, which closely matches the wake-rake integral value of $C_{d,min} = 0.020$ at $\alpha = -2^\circ$. Because XFOIL evaluates profile drag via the Squire-Young wake momentum deficit, it accounts for both skin friction and viscous form drag, mirroring the physical mechanism captured by the wake rake. In contrast, the surface pressure integration method diverges significantly from both theory and wake measurements. In the attached regime, pressure integration omits skin friction and suffers from tap discretisation, giving unphysically small and at points negative drag values, while beyond $\alpha = 10^\circ$ it surges well above both XFOIL and the wake data.

Finally, XFOIL confirms a flat pre-stall moment centred at $C_{m,c/4} \approx -0.015$. The experimental moment is similarly flat but shifted positive ($C_{m,c/4} \approx +0.01$ to $+0.06$) because lower-surface taps were not present aft of $x/c = 0.77$, missing key nose-down restoring pressure. Both experimental and theoretical data show a sudden positive moment increase at stall as the suction peak collapses and the centre of pressure moves aft.

Appendix C Analysis Code

All processing, from the raw data files to every figure, table and number quoted in the report, is performed by the single script below.

```

1  """AERO3260 Laboratory Report 1 - Analysis."""
2
3  import os
4
5  import numpy as np
6
7  OUT = os.path.join(os.path.dirname(__file__), "..", "Images")
8
9  R_UNIVERSAL = 8.3145          # J/(mol K)
10 M_AIR = 28.9645e-3           # kg/mol, dry air
11 R = R_UNIVERSAL / M_AIR      # = 287.058 J/(kg K)
12
13 MU_0 = 1.716e-5              # kg/(m s)
14 T_0 = 273.15                 # K
15 S = 110.4                    # K
16
17 T_CELSIUS = 23.0             # measured
18 P_ATM_HPA = 1031.7           # measured
19
20
21 # ----- Task 0 -----
22 def density(p_atm, T):
23     return M_AIR * p_atm / (R_UNIVERSAL * T)
24
25
26 def viscosity(T):
27     return MU_0 * (T / T_0) ** 1.5 * (T_0 + S) / (T + S)
28
29
30 def ambient(T_celsius=T_CELSIUS, p_hPa=P_ATM_HPA):
31     T = T_celsius + 273.15
32     p_atm = p_hPa * 100.0
33     return T, p_atm, density(p_atm, T), viscosity(T)
34
35
36 def reynolds(rho, U, L, mu):
37     return rho * U * L / mu
38
39
40 def dynamic_pressure(rho, U):
41     return 0.5 * rho * U ** 2
42
43
44 # ----- Task 1 -----
45 def cylinder_forces(theta, cp):
46     # Cd = (1/2) int Cp cos(th) dth, Cl = -(1/2) int Cp sin(th) dth
47     i = np.argsort(theta)
48     th, c = theta[i], cp[i]
49     th = np.append(th, th[0] + 2 * np.pi)
50     c = np.append(c, c[0])
51     return (-0.5 * np.trapezoid(c * np.sin(th), th),
52           0.5 * np.trapezoid(c * np.cos(th), th))
53
54
55 D_CYL, L_CYL = 0.050, 0.110
56 D_SUP, L_SUP = 0.025, 0.070
57

```

```

58 # Ports 1-13 alternate sides of the cylinder, so the taps close a contour on
59 # their own. Integration uses the signed angles, the Cp figure the magnitude.
60 PORT_ANGLES_SIGNED = np.array([0.0, 15.0, -30.0, 45.0, -60.0, 75.0, -90.0,
61                               105.0, -120.0, 135.0, -150.0, 165.0, 180.0])
62 PORT_ANGLES = np.abs(PORT_ANGLES_SIGNED) # ports 1-13
63 MANO = {6.0: np.array([386, 386, 388, 390, 393, 394, 393,
64                        394, 394, 394, 394, 394, 395.0]),
65         9.0: np.array([386, 387, 391, 393, 397, 397, 396,
66                        396, 397, 398, 397, 397, 398.0])}
67 MANO_STATIC = {6.0: 388.0, 9.0: 394.0} # port 14, tunnel static
68 PA_PER_MMH20 = 9.81 # readings are in mm H2O
69
70
71 def cp_manometer(h, h_static, q):
72     # lower reading = higher pressure, so  $p - p_{inf} = k (h_{static} - h)$ 
73     return PA_PER_MMH20 * (h_static - h) / q
74
75
76 def cylinder_cp(V, rho=None):
77     if rho is None:
78         rho = ambient()[2]
79     q = dynamic_pressure(rho, V)
80     return PORT_ANGLES, cp_manometer(MANO[V], MANO_STATIC[V], q)
81
82
83 def cylinder_coefficients(V, rho=None):
84     # no symmetry imposed: the lift comes from the data, not an assumption
85     _, cp = cylinder_cp(V, rho)
86     return cylinder_forces(np.deg2rad(PORT_ANGLES_SIGNED % 360.0), cp)
87
88
89 def support_drag(rho, V, cd_support):
90     return cd_support * dynamic_pressure(rho, V) * D_SUP * L_SUP
91
92
93 def cylinder_drag_from_balance(F_measured, rho, V, cd_support):
94     F = F_measured - support_drag(rho, V, cd_support)
95     return F / (dynamic_pressure(rho, V) * D_CYL * L_CYL)
96
97
98 def plot_cylinder_cp(rho, mu, fname="t1-cp.pdf"):
99     import usyd_plotstyle as ups
100     import matplotlib.pyplot as plt
101
102     fig, ax = plt.subplots(figsize=ups.figsize(1.0, aspect=0.46))
103     th = np.linspace(0.0, 180.0, 400)
104     ups.theory(ax, th, 1.0 - 4.0 * np.sin(np.deg2rad(th)) ** 2,
105              label=r"$C_p = 1-4\sin^2\theta$")
106     for i, V in enumerate((6.0, 9.0)):
107         a, cp = cylinder_cp(V, rho)
108         # +-0.5 mmH20 reading resolution, in Cp units; markers kept small so
109         # the +-0.10 bar at 9 m/s still shows past the symbol
110         dcp = PA_PER_MMH20 * 0.5 / dynamic_pressure(rho, V)
111         ups.measured(ax, a, cp, yerr=dcp, i=i, marker="o", ms=3.6, mew=0.5,
112                    label=rf"$U_{\infty} = \{V:.0f\}$ m/s",
113                    clip_on=False, zorder=5)
114     ax.grid(True, color="0.80", lw=0.5)
115     ax.set_axisbelow(True)
116     ax.axhline(0.0, color="0.6", lw=0.7, zorder=1)
117     ax.set_xlim(-5, 185)
118     ax.set_ylim(-3.6, 2.6) # headroom for the in-axes legend
119     ax.set_xticks(np.arange(0, 181, 30))

```

```

120 ax.set_yticks(np.arange(-3, 3, 1))
121 ups.finish(ax, r"$\theta$ (deg)", r"$C_p$")
122 # legend sits in the empty band above the mid-angle data
123 ax.legend(loc="upper center", bbox_to_anchor=(0.5, 1.0), ncol=3,
124          columns=1.2, handletextpad=0.5,
125          frameon=True, framealpha=1.0, edgecolor="0.75",
126          fancybox=False, borderpad=0.35)
127 fig.savefig(os.path.join(OUT, fname))
128 plt.close(fig)
129
130
131 # ----- Task 2 -----
132 # NACA 23012, c = 175 mm, Scanivalve MPS4264 at 16 angles of attack.
133 # Channels: 1-13 lower taps, 14-26 upper taps, 33-47 wake rake (8 mm pitch),
134 # 48 tunnel-inlet static, 49 side-wall static. Pressures are gauge against
135 # the scanner's atmospheric reference, which the two wind-off runs confirm.
136
137 CHORD = 0.175
138 DATA_DIR = os.path.join(os.path.dirname(__file__), "..", "..", "Wed 11AM Data")
139 ALPHAS = list(range(-10, 21, 2))
140
141 XC_LOW = np.array([.017, .037, .067, .10, .13, .19, .22, .27,
142                   .33, .44, .56, .67, .77])
143 XC_UP = np.array([.034, .059, .078, .11, .17, .21, .28, .31,
144                   .40, .52, .68, .79, 1.00])
145 RAKE_PITCH = 0.008
146 RAKE_Y = (np.arange(15) - 7.0) * RAKE_PITCH # centred on the middle port
147
148 # NACA 23012: 230 camber line (m = 0.2025, k1 = 15.957) and t = 0.12.
149 NACA_M, NACA_K1, NACA_T = 0.2025, 15.957, 0.12
150
151
152 def naca_camber(x):
153     m, k1 = NACA_M, NACA_K1
154     yc = np.where(x <= m,
155                  k1 / 6.0 * (x**3 - 3*m*x**2 + m**2*(3-m)*x),
156                  k1 / 6.0 * m**3 * (1.0 - x))
157     dyc = np.where(x <= m,
158                  k1 / 6.0 * (3*x**2 - 6*m*x + m**2*(3-m)),
159                  -k1 / 6.0 * m**3)
160     return yc, dyc
161
162
163 def naca_thickness(x):
164     t = NACA_T
165     yt = t/0.2 * (0.2969*np.sqrt(x) - 0.1260*x - 0.3516*x**2
166                  + 0.2843*x**3 - 0.1015*x**4)
167     dyt = t/0.2 * (0.14845/np.sqrt(np.maximum(x, 1e-9)) - 0.1260
168                  - 0.7032*x + 0.8529*x**2 - 0.4060*x**3)
169     return yt, dyt
170
171
172 def surface(x, upper):
173     yc, dyc = naca_camber(x)
174     yt, dyt = naca_thickness(x)
175     if upper:
176         return yc + yt, dyc + dyt
177     return yc - yt, dyc - dyt
178
179
180 def load_run(alpha):
181     name = f"N{-alpha}.csv" if alpha < 0 else f"{alpha}.csv"

```

```

182     a = np.genfromtxt(os.path.join(DATA_DIR, name), delimiter=",",
183                       skip_header=1)
184     return a[:, 2:66]      # columns SN2103-1 .. SN2103-64
185
186
187 def run_stats(alpha):
188     cols = load_run(alpha)
189     mean = np.nanmean(cols, axis=0)
190     sd = np.nanstd(cols, axis=0)
191     n = np.sum(~np.isnan(cols[:, 0]))
192     return mean, sd, n
193
194
195 U_NOMINAL = 12.0      # tunnel set speed for the aerofoil runs, m/s
196
197
198 def freestream():
199     """Freestream total pressure per run, and the reference dynamic pressure.
200
201     p0 is the mean of the two cleanest (largest) rake totals of each run and
202     is steady to about 1 Pa across all sixteen runs.
203
204     q is referenced to the nominal tunnel speed rather than to the
205     rake-total minus wall-static difference, since the wall taps sit in the
206     model's own pressure field and shift by up to 45 Pa at high lift.
207     freestream_measured_q() returns the measured difference for comparison.
208     """
209     p0s = {}
210     for a in ALPHAS:
211         mean, _, _ = run_stats(a)
212         p0s[a] = float(np.mean(np.sort(mean[32:47])[-2:]))
213     rho = ambient()[2]
214     return p0s, float(0.5 * rho * U_NOMINAL ** 2)
215
216
217 def freestream_measured_q():
218     """Rake-total minus side-wall static, averaged over |alpha| <= 4 deg."""
219     qs = []
220     for a in ALPHAS:
221         if abs(a) <= 4:
222             mean, _, _ = run_stats(a)
223             qs.append(np.mean(np.sort(mean[32:47])[-2:]) - mean[48])
224     return float(np.mean(qs))
225
226
227 def aerofoil_cp(alpha, p0s, q):
228     mean, sd, n = run_stats(alpha)
229     p_inf = p0s[alpha] - q
230     cp_low = (mean[0:13] - p_inf) / q
231     cp_up = (mean[13:26] - p_inf) / q
232     ci_low = 1.96 * sd[0:13] / np.sqrt(n) / q
233     ci_up = 1.96 * sd[13:26] / np.sqrt(n) / q
234     return cp_low, cp_up, ci_low, ci_up
235
236
237 def aerofoil_coefficients(alpha, p0s, q):
238     """Cn, Ca and Cm_c/4 by the trapezoidal rule over the tap locations
239     (equations 3 and 4 of the brief), then rotated to Cl and Cd."""
240     cpl, cpu, _, _ = aerofoil_cp(alpha, p0s, q)
241     _, dyl = surface(XC_LOW, upper=False)
242     yl, _ = surface(XC_LOW, upper=False)
243     yu, dyu = surface(XC_UP, upper=True)

```

```

244     cn = np.trapezoid(cpl, XC_LOW) - np.trapezoid(cpu, XC_UP)
245     ca = np.trapezoid(cpu * dyu, XC_UP) - np.trapezoid(cpl * dyl, XC_LOW)
246     cm = (np.trapezoid(cpu * (0.25 - XC_UP), XC_UP)
247           - np.trapezoid(cpl * (0.25 - XC_LOW), XC_LOW)
248           + np.trapezoid(cpu * dyu * yu, XC_UP)
249           - np.trapezoid(cpl * dyl * yl, XC_LOW))
250     ar = np.deg2rad(alpha)
251     cl = cn * np.cos(ar) - ca * np.sin(ar)
252     cd = cn * np.sin(ar) + ca * np.cos(ar)
253     return cn, ca, cm, cl, cd
254
255
256 def wake_profile(alpha, p0s, q):
257     mean, sd, n = run_stats(alpha)
258     p_inf = p0s[alpha] - q
259     ratio = np.clip((mean[32:47] - p_inf) / q, 0.0, None)
260     u = np.sqrt(ratio)
261     # r.m.s. pressure fluctuation carried through the square root; d(u/U)/dp
262     # diverges as u -> 0, so the bar is dropped below u/U = 0.3
263     err = np.where(u > 0.30, (sd[32:47] / q) / np.maximum(2 * u, 0.6), 0.0)
264     return u, err
265
266
267 def wake_drag(alpha, p0s, q):
268     u, _ = wake_profile(alpha, p0s, q)
269     integrand = u * (1.0 - u)
270     return 2.0 / CHORD * np.trapezoid(integrand, RAKE_Y)
271
272
273 def plot_aerofoil_cp(p0s, q, fname="t2-cp.pdf"):
274     """Negative and positive incidence side by side, as the brief suggests.
275     Colour encodes alpha on one shared diverging scale, filled circles the
276     upper surface and open squares the lower. Bars are the r.m.s.
277     fluctuation over the 2500 frames, not the confidence interval on the
278     mean, which is smaller than a marker at every port."""
279     import usyd_plotstyle as ups
280     import matplotlib.pyplot as plt
281     from matplotlib import cm, colors
282
283     fig, axes = plt.subplots(1, 2, figsize=ups.figsize(1.0, aspect=0.395),
284                             sharey=True)
285     norm = colors.TwoSlopeNorm(vmin=min(ALPHAS), vcenter=0.0,
286                               vmax=max(ALPHAS))
287     # Diverging about alpha = 0 with a neutral grey midpoint: a white centre
288     # would make the mid-range angles vanish against the page.
289     cmap = colors.LinearSegmentedColormap.from_list(
290         "usyd_div", [ups.COLOURS[0], "#8A8A8A", ups.COLOURS[1]])
291     groups = [(axes[0], [a for a in ALPHAS if a < 0], r"$\alpha < 0$"),
292              (axes[1], [a for a in ALPHAS if a >= 0], r"$\alpha \geq 0$")]
293     for ax, subset, title in groups:
294         for a in subset:
295             cpl, cpu, _, _ = aerofoil_cp(a, p0s, q)
296             _, sd, n = run_stats(a)
297             c = cmap(norm(a))
298             ax.errorbar(XC_UP, cpu, yerr=sd[13:26] / q, color=c, lw=0.8,
299                        marker="o", ms=3.0, mfc=c, mec="k", mew=0.3,
300                        elinewidth=0.5, capsize=1.0, zorder=3)
301             ax.errorbar(XC_LOW, cpl, yerr=sd[0:13] / q, color=c, lw=0.8,
302                        ls="--", marker="s", ms=3.0, mfc="none", mec=c,
303                        mew=0.8, elinewidth=0.5, capsize=1.0, zorder=3)
304             ax.axhline(0.0, color="0.75", lw=0.6, zorder=0)
305             ax.grid(True, color="0.88", lw=0.4)

```



```

306     ax.set_axisbelow(True)
307     ax.set_xlim(-0.02, 1.02)
308     ax.set_xlabel(r"$x/c$")
309     ax.set_title(title, fontsize=9, pad=3)
310 axes[0].set_ylim(1.35, -5.0)           # inverted: suction upward
311 axes[0].set_ylabel(r"$C_p$")
312 # marker key, kept separate from the colour scale. Drawn as real
313 # errorbar containers so the legend shows the bars themselves.
314 key = [axes[0].errorbar([], [], yerr=[], color="0.3", lw=0.8, marker="o",
315                        ms=3.0, mfc="0.3", mec="k", mew=0.3,
316                        elinewidth=0.5, capsize=1.6,
317                        label="Upper surface"),
318        axes[0].errorbar([], [], yerr=[], color="0.3", lw=0.8, ls="--",
319                        marker="s", ms=3.0, mfc="none", mec="0.3",
320                        mew=0.8, elinewidth=0.5, capsize=1.6,
321                        label="Lower surface")]
322 axes[0].legend(handles=key, fontsize=7, loc="upper right", frameon=True,
323               framealpha=1.0, edgecolor="0.75", fancybox=False,
324               borderpad=0.3, handlelength=1.6)
325 fig.subplots_adjust(wspace=0.13)
326 cb = fig.colorbar(cm.ScalarMappable(norm=norm, cmap=cmap), ax=axes,
327                  pad=0.012, fraction=0.028, ticks=ALPHAS)
328 cb.set_label(r"$\alpha$ (deg)", fontsize=8)
329 cb.ax.tick_params(labelsize=7)
330 fig.savefig(os.path.join(OUT, fname))
331 plt.close(fig)
332
333
334 def plot_aerofoil_cp_grid(p0s, q, fname="t2-cp-grid.pdf"):
335     """Per-angle breakdown of the same data as t2-cp.pdf, for the appendix
336     where there is no page limit and the panels can be read individually."""
337     import usyd_plotstyle as ups
338     import matplotlib.pyplot as plt
339
340     fig, axes = plt.subplots(4, 4, figsize=ups.figsize(1.0, aspect=0.92),
341                             sharex=True, sharey=True)
342     for k, (a, ax) in enumerate(zip(ALPHAS, axes.flat)):
343         cpl, cpu, _, _ = aerofoil_cp(a, p0s, q)
344         _, sd, n = run_stats(a)
345         ax.errorbar(XC_UP, cpu, yerr=sd[13:26] / q, marker="o", ms=2.8,
346                   mfc=ups.COLOURS[0], mec="k", mew=0.4,
347                   color=ups.COLOURS[0], lw=0.8, elinewidth=0.5,
348                   capsize=1.0, label="Upper")
349         ax.errorbar(XC_LOW, cpl, yerr=sd[0:13] / q, marker="s", ms=2.8,
350                   mfc="none", mec=ups.COLOURS[1], mew=0.7,
351                   color=ups.COLOURS[1], lw=0.8, ls="--", elinewidth=0.5,
352                   capsize=1.0, label="Lower")
353         ax.axhline(0.0, color="0.75", lw=0.5, zorder=0)
354         ax.set_title(rf"$\alpha = {a}^\circ$", fontsize=8, pad=2.5)
355         ax.tick_params(labelsize=7)
356         ax.grid(True, color="0.88", lw=0.4)
357         ax.set_axisbelow(True)
358     axes[0, 0].set_ylim(1.4, -5.2)
359     for ax in axes[-1, :]:
360         ax.set_xlabel(r"$x/c$", fontsize=8)
361     for ax in axes[:, 0]:
362         ax.set_ylabel(r"$C_p$", fontsize=8)
363     handles, labels = axes[0, 0].get_legend_handles_labels()
364     fig.legend(handles, labels, fontsize=8, ncol=2, loc="lower center",
365               frameon=False, bbox_to_anchor=(0.5, -0.005))
366     fig.subplots_adjust(hspace=0.28, wspace=0.10, bottom=0.10)
367     fig.savefig(os.path.join(OUT, fname))

```

```

368 plt.close(fig)
369
370
371 def plot_aerofoil_polars(p0s, q, fname="t2-polars.pdf"):
372     import usyd_plotstyle as ups
373     import matplotlib.pyplot as plt
374
375     res = np.array([aerofoil_coefficients(a, p0s, q) for a in ALPHAS])
376     cl, cd, cm = res[:, 3], res[:, 4], res[:, 2]
377     cdw = np.array([wake_drag(a, p0s, q) for a in ALPHAS])
378     al = np.array(ALPHAS)
379     fig, axgrid = plt.subplots(2, 2, figsize=ups.figsize(1.0, aspect=0.315))
380     # points only, except the Cd panel where a line separates the two sets
381     panels = [(axgrid[0,0], al, cl, r"$\alpha$ (deg)", r"$C_l$", 0.0),
382              (axgrid[0,1], al, cd, r"$\alpha$ (deg)", r"$C_d$", 0.7),
383              (axgrid[1,0], al, cm, r"$\alpha$ (deg)", r"$C_{m,c/4}$", 0.0),
384              (axgrid[1,1], cd, cl, r"$C_d$", r"$C_l$", 0.0)]
385     for ax, x, y, xl, yl, lw in panels:
386         ax.plot(x, y, marker="o", ms=2.9, mfc=ups.COLOURS[0], mec="k",
387                mew=0.45, color=ups.COLOURS[0], lw=lw,
388                ls="-" if lw else "none")
389         ax.grid(True, color="0.85", lw=0.5)
390         ax.set_axisbelow(True)
391         ax.set_xlabel(xl, fontsize=9, labelpad=1.5)
392         ax.set_ylabel(yl, fontsize=9, labelpad=1.5)
393         ax.tick_params(labelsize=8)
394     axgrid[0,0].axhline(0.0, color="0.7", lw=0.6, zorder=1)
395     axgrid[1,0].axhline(0.0, color="0.7", lw=0.6, zorder=1)
396     # thin aerofoil slope for reference on the lift curve
397     aa = np.linspace(-8, 12, 50)
398     axgrid[0,0].plot(aa, 2*np.pi*np.deg2rad(aa - (-1.97)), color="k", lw=0.9,
399                     zorder=1, label=r"$2\pi(\alpha-\alpha_0)$")
400     axgrid[0,0].legend(fontsize=7.5, loc="upper left", frameon=True,
401                       framealpha=1.0, edgecolor="0.75", fancybox=False,
402                       borderpad=0.3, handlelength=1.2)
403     # wake-integral drag on the same Cd panel (task 2.4 comparison)
404     axgrid[0,1].plot(al, cdw, marker="s", ms=2.7, mfc="none",
405                     mec=ups.COLOURS[1], mew=0.7, color=ups.COLOURS[1],
406                     lw=0.7, ls="--", label="Wake")
407     axgrid[0,1].plot([], [], marker="o", ms=2.7, mfc=ups.COLOURS[0], mec="k",
408                     mew=0.45, color=ups.COLOURS[0], lw=0.7, label="Pressure")
409     handles, labels = axgrid[0,1].get_legend_handles_labels()
410     axgrid[0,1].legend(handles[::1], labels[::1], fontsize=7.5,
411                       loc="upper left", frameon=True, framealpha=1.0,
412                       edgecolor="0.75", fancybox=False, borderpad=0.3,
413                       handlelength=1.4)
414     for ax in axgrid.flat:
415         ax.tick_params(labelsize=8)
416         ax.xaxis.label.set_size(9)
417         ax.yaxis.label.set_size(9)
418     fig.subplots_adjust(wspace=0.20, hspace=0.45, left=0.075, right=0.985, top=0.985, bottom=0.115)
419     fig.savefig(os.path.join(OUT, fname))
420     plt.close(fig)
421
422
423 def plot_wake(p0s, q, fname="t2-wake.pdf"):
424     """Waterfall of normalised wake profiles, eight angles of attack per row."""
425     import usyd_plotstyle as ups
426     import matplotlib.pyplot as plt
427
428     per_row = 8
429     step = 1.30

```

```

430 col = ups.COLOURS[0]
431 y = RAKE_Y / CHORD
432
433 fig, axes = plt.subplots(2, 1, figsize=ups.figsize(1.0, aspect=0.435),
434                          sharey=True)
435 for ax, row in zip(axes, (ALPHAS[:per_row], ALPHAS[per_row:])):
436     ticks, labels = [], []
437     for k, a in enumerate(row):
438         u, ci = wake_profile(a, p0s, q)
439         # truncated at zero: the velocity cannot go negative
440         bar = np.vstack([np.minimum(ci, u), ci])
441         x0 = k * step
442         # each trace carries its own u/U = 1 datum, read vertically
443         ax.axvline(x0 + 1.0, color="0.86", lw=0.6, zorder=0)
444         ax.fill_betweenx(y, x0 + u, x0 + 1.0, color=col, alpha=0.16,
445                          lw=0, zorder=1)
446         ax.errorbar(x0 + u, y, xerr=bar, marker="o", ms=2.6, mfc=col,
447                     mec="k", mew=0.35, color=col, lw=0.8, elinewidth=0.5,
448                     capsize=1.0, zorder=2)
449         ax.text(x0 + 0.5, 1.04, rf"${a}^\circ$", fontsize=7.5, ha="center",
450                 va="bottom", transform=ax.get_xaxis_transform())
451         ticks += [x0, x0 + 1.0]
452         labels += ["0", "1"]
453     ax.axhline(0.0, color="0.80", ls=(0, (1, 2.5)), lw=0.6, zorder=0)
454     ax.set_xticks(ticks)
455     ax.set_xticklabels(labels, fontsize=6.5)
456     ax.set_xlim(-0.30, (per_row - 1) * step + 1.32)
457     ax.set_ylim(-0.40, 0.40)
458     ax.set_yticks([-0.3, 0.0, 0.3])
459     ax.set_ylabel(r"$y/c$", fontsize=9)
460     ax.tick_params(axis="y", labelsize=8)
461     ax.tick_params(axis="x", top=False)
462 axes[1].set_xlabel(r"$u/U_\infty$, offset for each angle of attack",
463                   fontsize=9)
464 fig.subplots_adjust(hspace=0.34)
465 fig.savefig(os.path.join(OUT, fname))
466 plt.close(fig)
467
468 def plot_cd_compare(p0s, q, fname="t2-cd.pdf"):
469     import usyd_plotstyle as ups
470     import matplotlib.pyplot as plt
471
472     res = np.array([aerofoil_coefficients(a, p0s, q) for a in ALPHAS])
473     cdw = np.array([wake_drag(a, p0s, q) for a in ALPHAS])
474     al = np.array(ALPHAS)
475     fig, ax = plt.subplots(figsize=ups.figsize(0.62, aspect=0.62))
476     ax.plot(al, res[:, 4], marker="o", ms=3.6, mfc=ups.COLOURS[0], mec="k",
477            mew=0.5, color=ups.COLOURS[0], lw=0.8, label="Surface pressure")
478     ax.plot(al, cdw, marker="s", ms=3.4, mfc="none", mec=ups.COLOURS[1],
479            mew=0.8, color=ups.COLOURS[1], lw=0.8, ls="--",
480            label="Wake integral")
481     ax.axhline(0.0, color="0.7", lw=0.6, zorder=1)
482     ax.grid(True, color="0.85", lw=0.5)
483     ax.set_axisbelow(True)
484     ax.set_xlabel(r"$\alpha$ (deg)", fontsize=9)
485     ax.set_ylabel(r"$C_d$", fontsize=9)
486     ax.tick_params(labelsize=8)
487     ax.legend(fontsize=8, loc="upper center", frameon=True, framealpha=1.0,
488              edgcolor="0.75", fancybox=False)
489     fig.savefig(os.path.join(OUT, fname))
490     plt.close(fig)
491

```

```

492
493 def write_raw_tables():
494     """Mean gauge pressures (Pa) for every used channel at every
495     angle of attack, for the raw-data appendix."""
496     tdir = os.path.join(os.path.dirname(__file__), '..', 'Tables')
497     os.makedirs(tdir, exist_ok=True)
498     means = {a: run_stats(a)[0] for a in ALPHAS}
499     nl = chr(10)
500     bs = chr(92)
501     head = ' & '.join('{ ' + str(a) + '}' for a in ALPHAS)
502     surf = [(str(i), i - 1) for i in range(1, 27)]
503     rake = [(str(i), i - 1) for i in range(33, 48)]
504             + [('48', 47), ('49', 48)])
505     for fname, rows in (('t2-raw-surface.tex', surf),
506                        ('t2-raw-rake.tex', rake)):
507         lines = ['% Generated by Code/lab1.py -- do not edit.',
508                 bs + 'begin{tabular}{l *{16}{S[table-format=-3.0]}}',
509                 ' ' + bs + 'toprule',
510                 ' Port & ' + head + ' ' + bs + bs,
511                 ' ' + bs + 'midrule']
512         for name, idx in rows:
513             vals = ' & '.join(format(means[a][idx], '.0f')
514                               for a in ALPHAS)
515             lines.append(' ' + name + ' & ' + vals + ' ' + bs + bs)
516         lines += [' ' + bs + 'bottomrule', bs + 'end{tabular}', '']
517         with open(os.path.join(tdir, fname), 'w') as f:
518             f.write(nl.join(lines))
519
520 if __name__ == "__main__":
521     T, p_atm, rho, mu = ambient()
522     print(f"R      = {R:.3f} J/(kg K)")
523     print(f"rho    = {rho:.3f} kg/m^3   mu = {mu:.3e} kg/(m s)")
524     for V in (6.0, 9.0):
525         th, cp = cylinder_cp(V, rho)
526         cl, cd = cylinder_coefficients(V, rho)
527         print(f"V = {V:.0f} m/s  Re = {reynolds(rho, V, D_CYL, mu):.3g} "
528               f"Cl = {cl:+.3f}  Cd = {cd:+.3f}")
529     plot_cylinder_cp(rho, mu)
530
531     p0s, q2 = freestream()
532     U2 = np.sqrt(2*q2/rho)
533     print(f"Task 2: q = {q2:.1f} Pa  U = {U2:.2f} m/s "
534           f"Re = {reynolds(rho, U2, CHORD, mu):.3g}")
535     print(" alpha  Cn      Ca      Cm4      Cl      Cd      Cd_wake")
536     for a in ALPHAS:
537         cn, ca, cm, cl, cd = aerofoil_coefficients(a, p0s, q2)
538         cdw = wake_drag(a, p0s, q2)
539         print(f" {a:5d} {cn:7.3f} {ca:7.3f} {cm:7.3f} "
540               f"{cl:7.3f} {cd:7.3f} {cdw:7.3f}")
541     plot_aerofoil_cp(p0s, q2)
542     plot_aerofoil_cp_grid(p0s, q2)
543     plot_aerofoil_polars(p0s, q2)
544     plot_wake(p0s, q2)
545     write_raw_tables()

```

Listing 1: Analysis code for the experimental ambient conditions, cylinder and aerofoil processing.